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A Spectrum of Diffusion Models for Hierarchical Control of Attention: From Sequential to Parallel Processing

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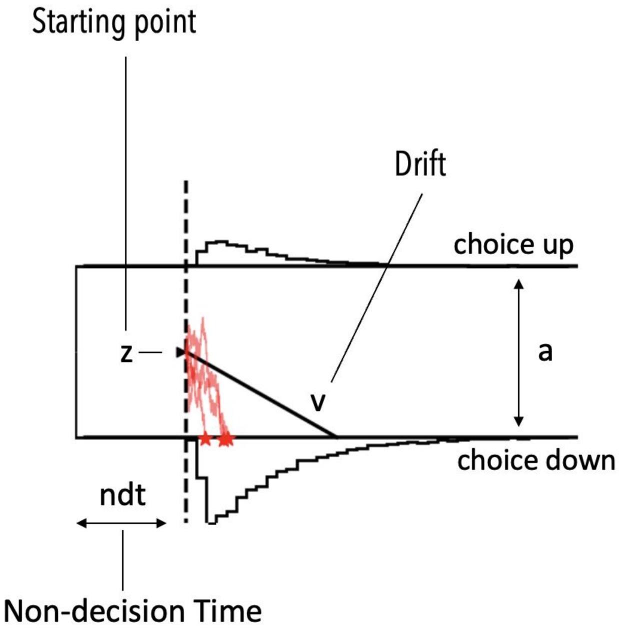
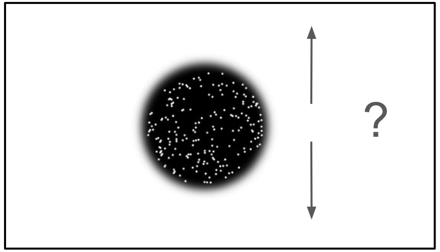
The beloved Drift Diffusion Model…

Very widely applied!





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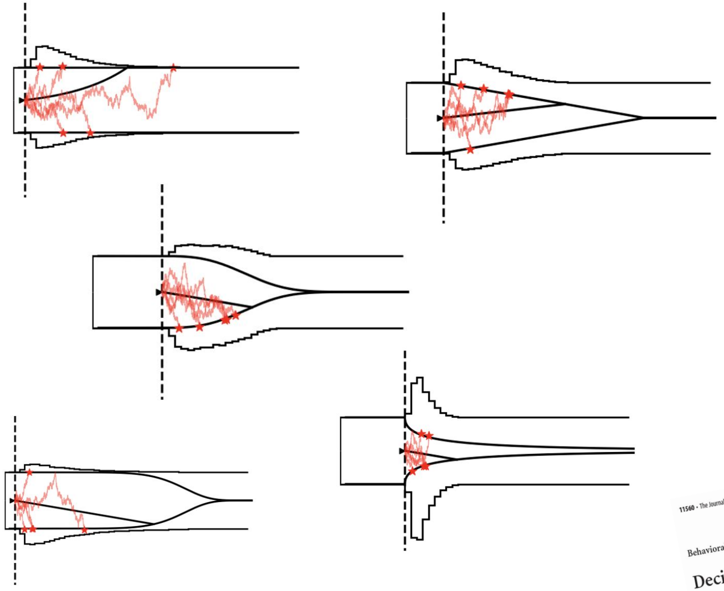
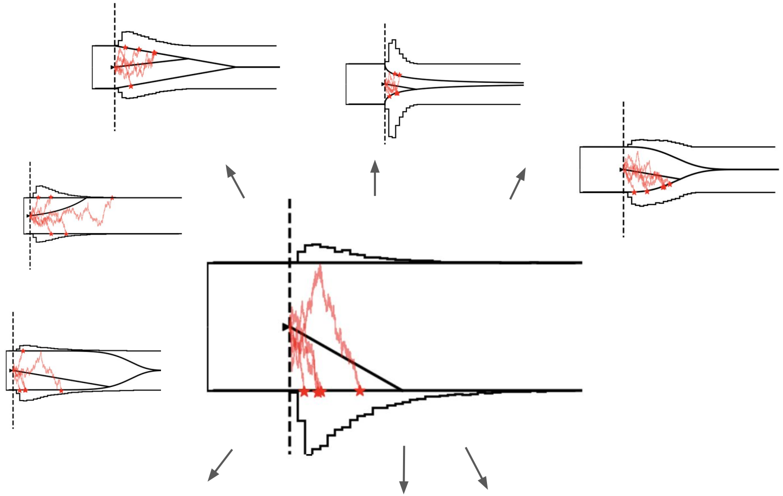


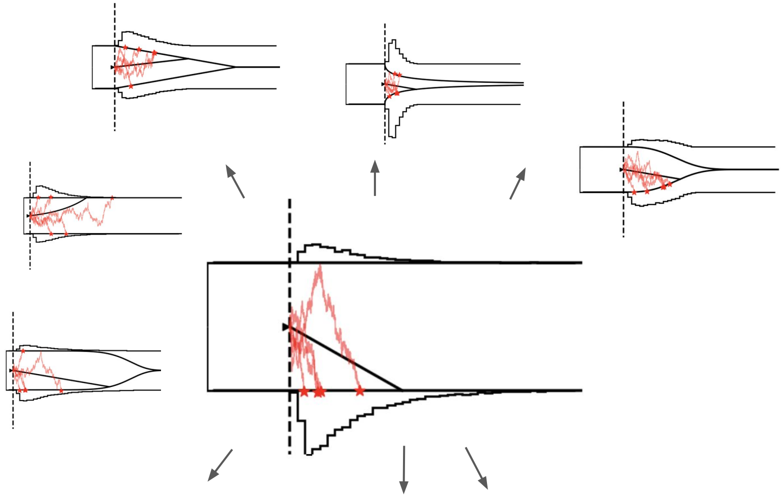
A host of **variations** on the basic drift diffusion model have been **motivated in the literature**.

Outsized influence of DDM



Until recently **empirical data analysis** has put an **outsized focus** on the basic **DDM**.

Outsized influence of DDM



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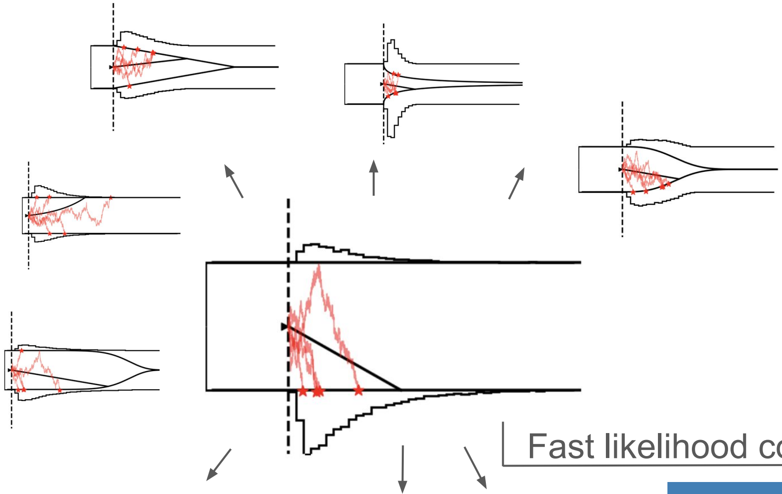
Why?

Outsized influence of DDM



Until recently **empirical data analysis** has put an **outsized focus** on the basic **DDM**.

Why?

Fast likelihood computations!

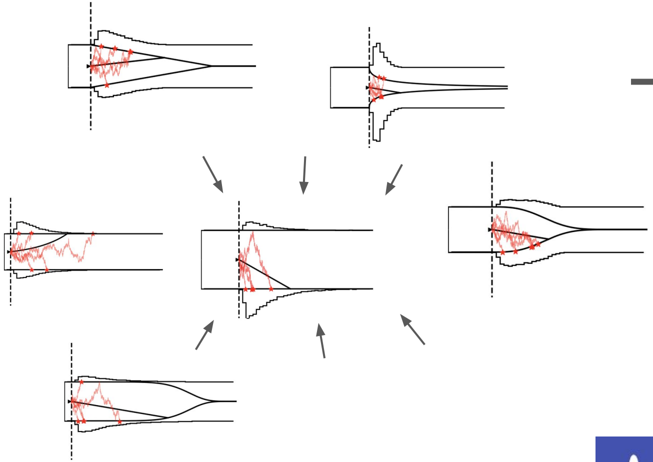
$$p_M(rt, c = 1 | v, a, z) = \frac{\pi}{a^2} \exp(-vaz - \frac{v^2 t}{2}) \sum_{k=1}^{\infty} k \exp\left(-\frac{k^2 \pi^2 t}{2a^2}\right) \sin(k\pi z)$$

$$p(\theta|x) \propto p(x|\theta) p(\theta)$$

Software Support!

- HDDM
- HSSM
- fast-dm

SBI / New Toolkit

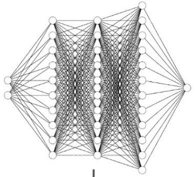


Recent **boost in methods** and software support, **leveled the playing field**!

How?

sbi

HSSM



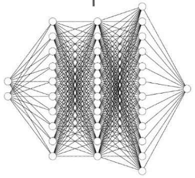
LANs
MNLEs
SNLE
...

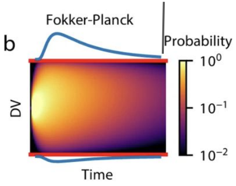
$$p(\theta|x) \propto p(x|\theta)p(\theta)$$

sbi

BayesFlow

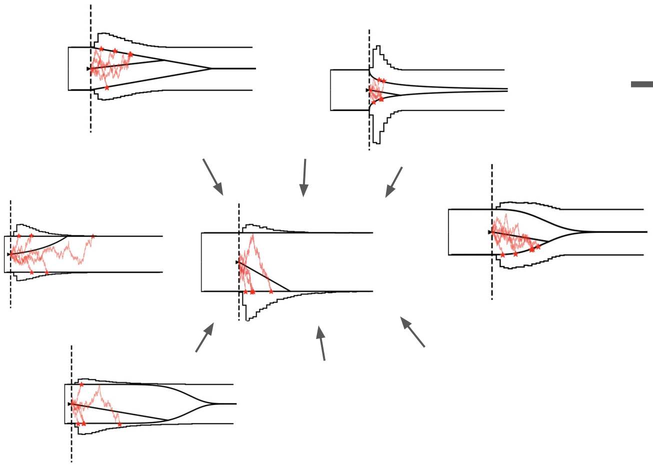
BayesFlow
SNPE
...





PyDDM

SBI / New Toolkit



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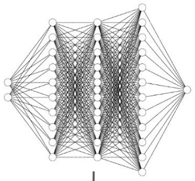

LANs
MNLEs
SNLE
...

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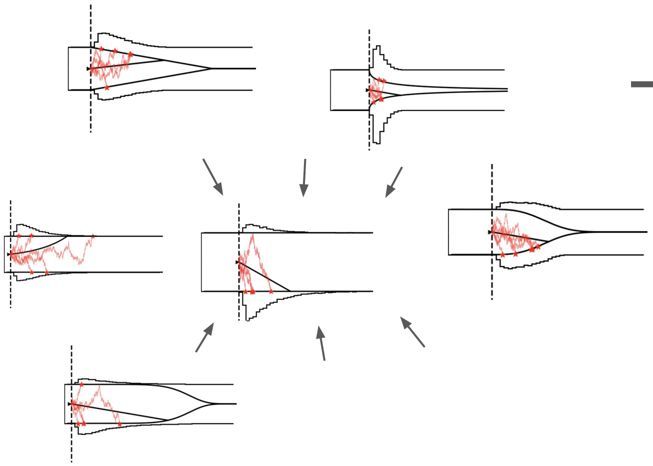
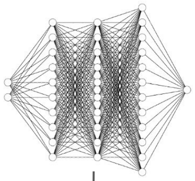




LANs
MNLEs
SNLE
…

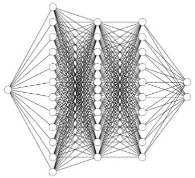
$$p(\theta|x) \propto p(x|\theta)p(\theta)$$

Key Idea:

Learn from simulations once → reuse

Small ad...

HSSM



LANs
MNLEs
SNLE
...

https://lnccbrown.github.io/HSSM/

Home Getting Started API References Tutorials Contributing

HOME

Overview
Credits
License
Changelog

Overview

HSSM

BRAIN STORM

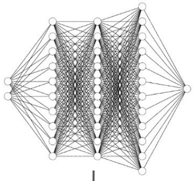
pypi v0.2.1 downloads 511/month python 3.10 | 3.11 pull requests 5 open build passing Stars 68
code style black

HSSM (Hierarchical Sequential Sampling Modeling) is a modern Python toolbox that provides state-of-the-art likelihood approximation methods within the Python Bayesian ecosystem. It facilitates hierarchical model building and inference via fast and robust MCMC samplers. User-friendly, extensible, and flexible, HSSM can rigorously estimate the impact of neural and other trial-by-trial covariates through parameter-wise mixed-effects models for a large variety of cognitive process models.

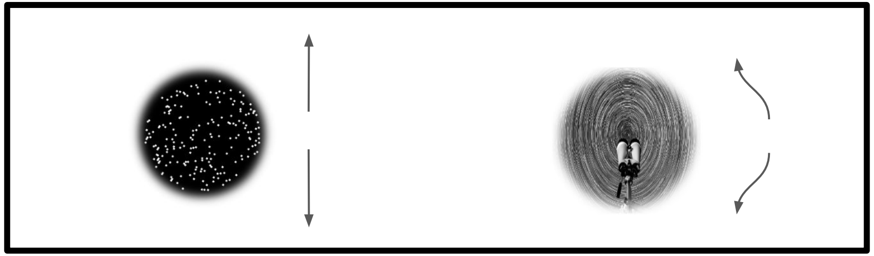
HSSM is a BRAINSTORM project in collaboration with the Center for Computation and Visualization (CCV) and the Center for Computational Brain Science within the Carney Institute at Brown University.

New horizons…

These new methods allow us to become more **ambitious** with our **computational models**, and in turn about respective **experimental designs**!



$$p(\theta|x) \propto p(x|\theta)p(\theta)$$



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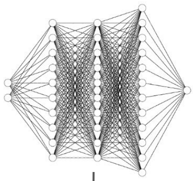

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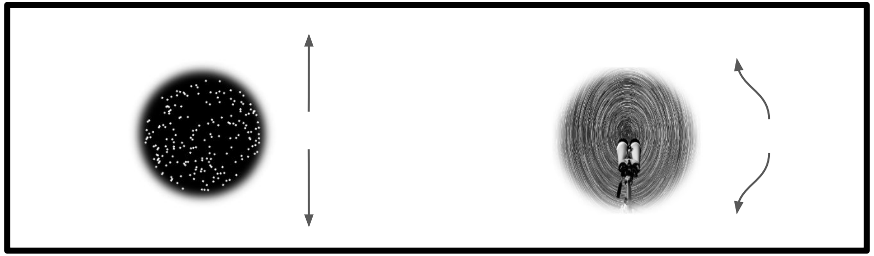
Re-analysis of complicated experiments via computational modeling



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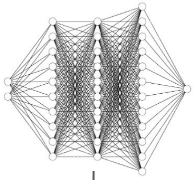

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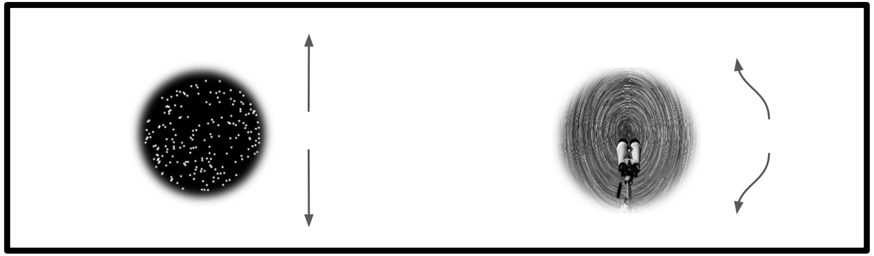
Re-analysis of complicated experiments via computational modeling

Influence the design of ***new experiments***



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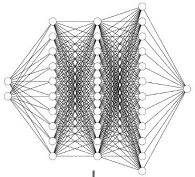
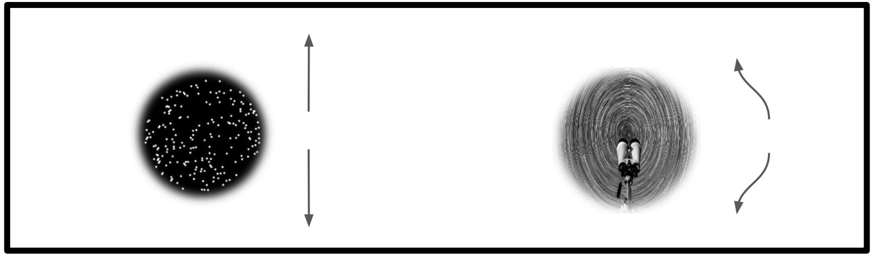


$$\color{blue}{p(\theta|x)} \propto \color{green}{p(x|\theta)}\color{red}{p(\theta)}$$

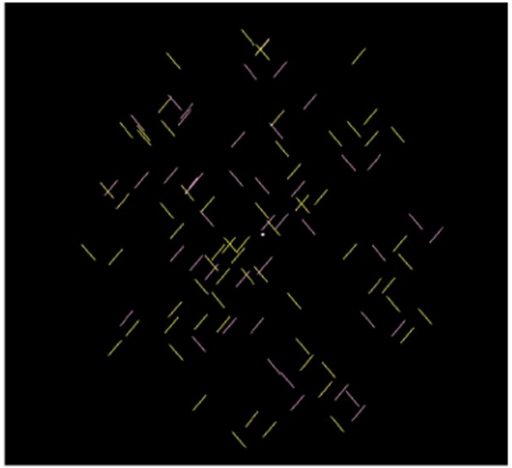
Re-analysis or complicated experiments via computational modeling

Influence the design of ***new experiments***

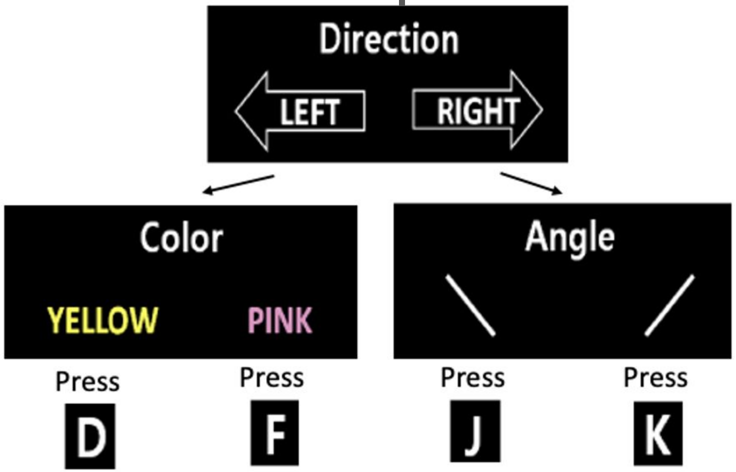


Setting...

Three features



Stimulus Screen

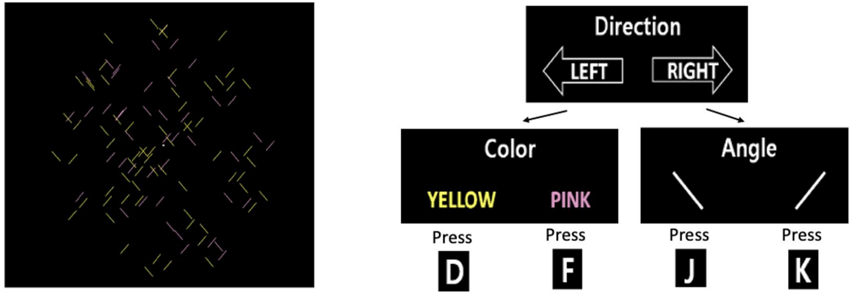


Task Instructions

Hierarchical Task Design

4 choice options

Hierarchical Decision Making



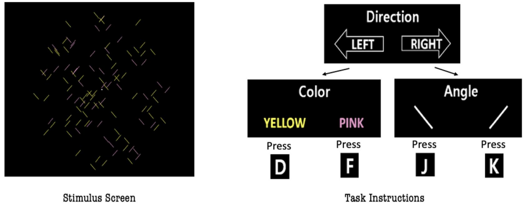
Stimulus Screen

Task Instructions

Overarching question:

Does the ***hierarchical task structure*** induce a ***sequential or parallel decision making*** process?

Hierarchical Decision Making



Overarching question:

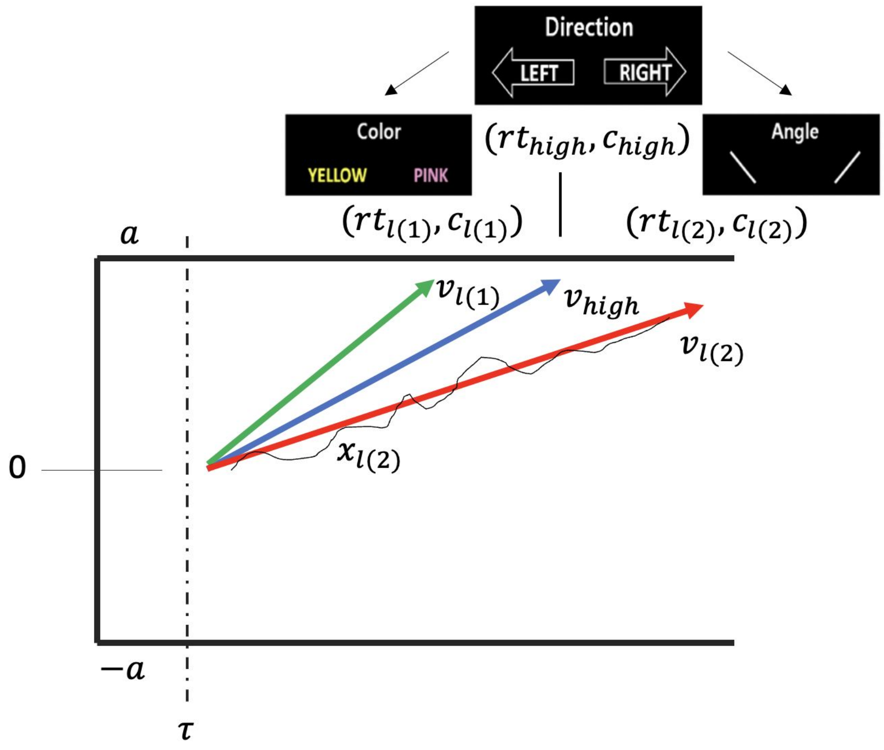
Does the ***hierarchical task structure*** induce a ***sequential or parallel decision making*** process?

The goal is to test three types of cognitive process models:

1. **Parallel** Processing
2. **Sequential** Processing
3. Resource Rational

Hybrids

Parallel Processing



choice is triggered after **both high dim** and **relevant low dim** processes are finished

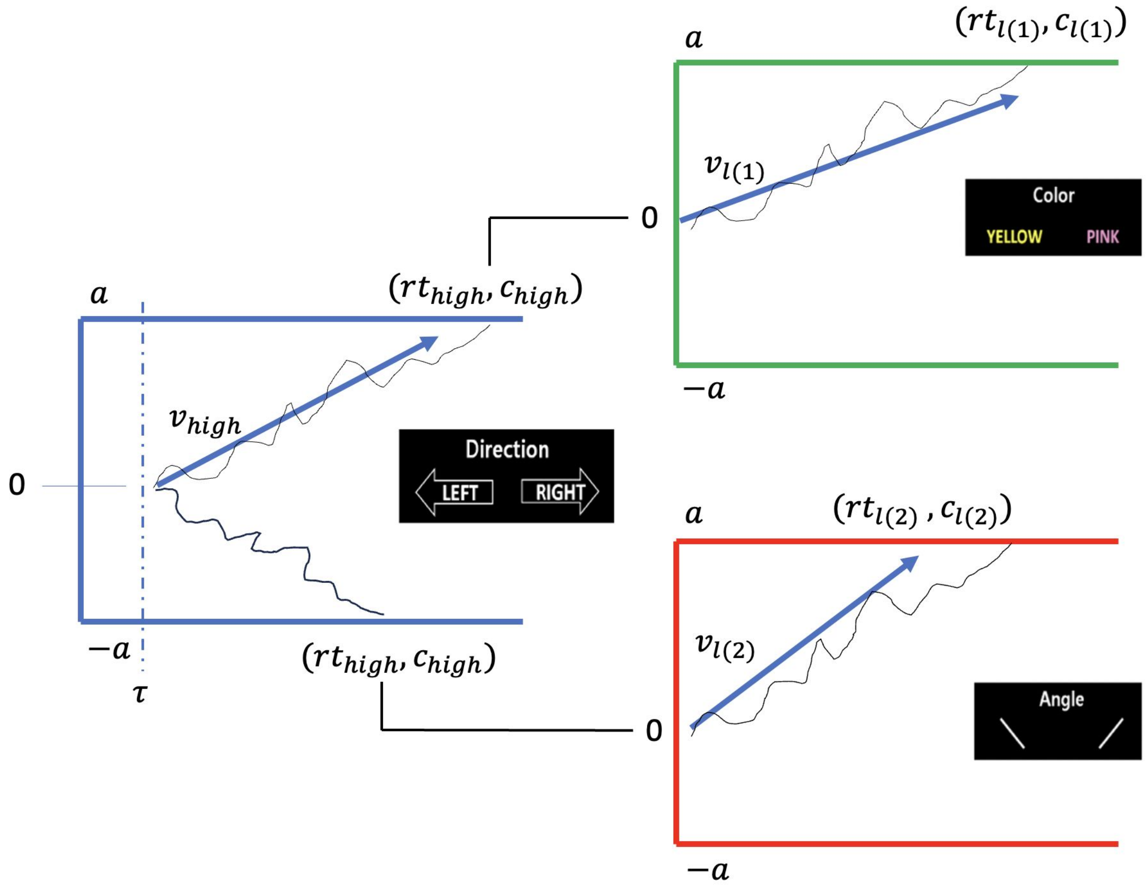
$$rt = \max(rt_{high}, rt_{l(c_h)})$$

$$c = (c_h, c_{l(c_h)})$$

high dim choice

relevant low dim choice

Sequential Processing



RT is the simple **sum** of **high dim** and **relevant low dim** process

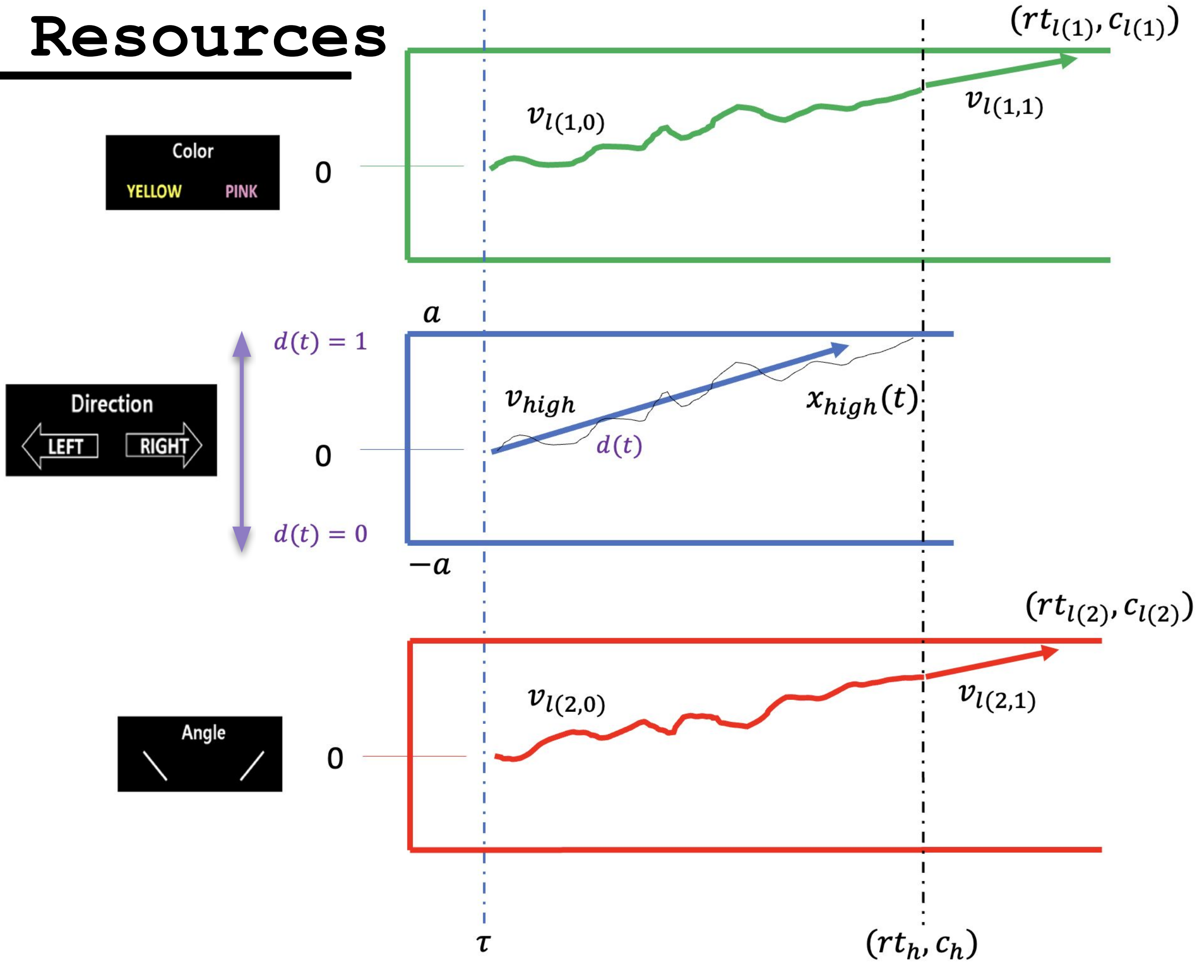
$$rt = rt_{high} + rt_{l(c_{high})}$$

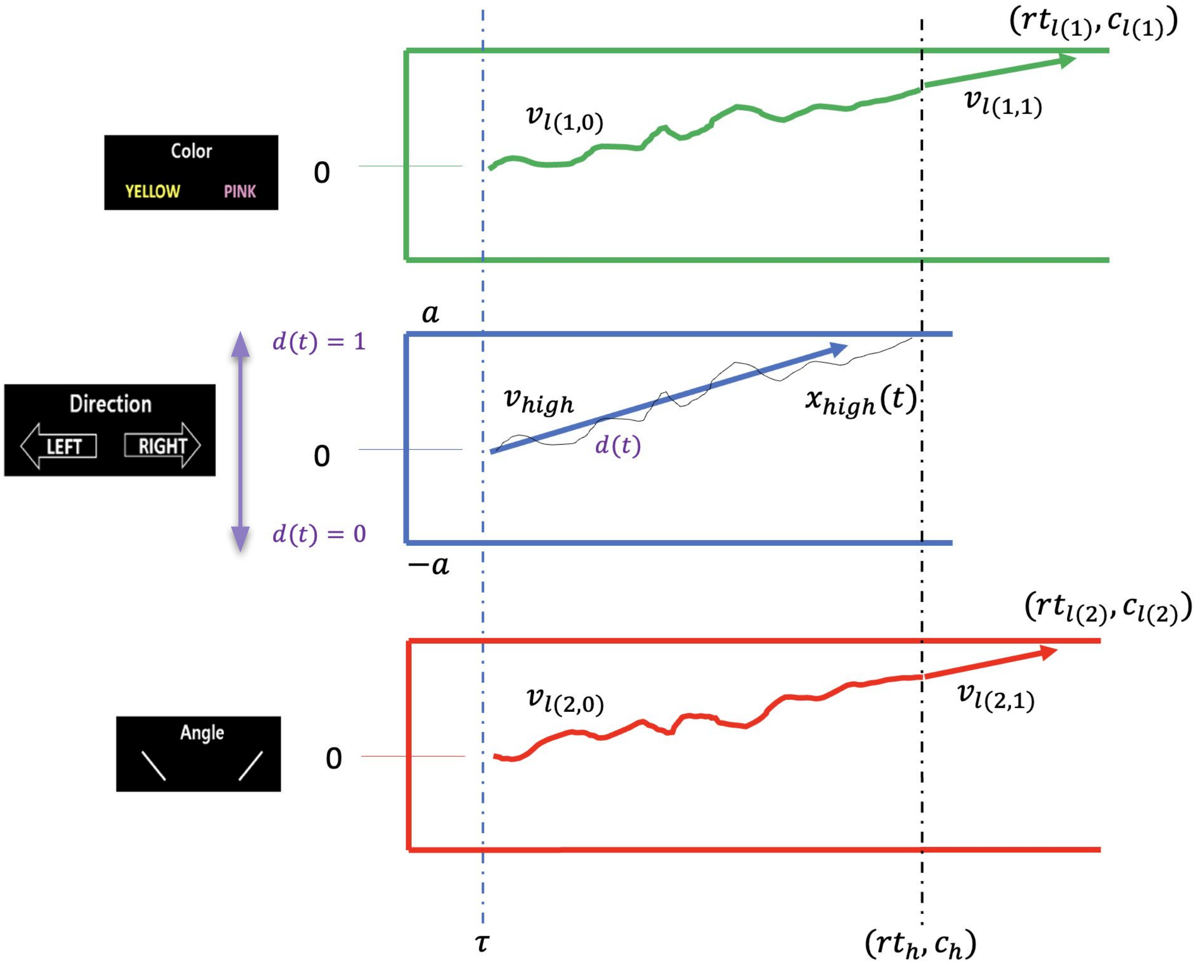
$$c = (c_{high}, c_{l(c_{high})})$$

high dim choice

relevant low dim choice

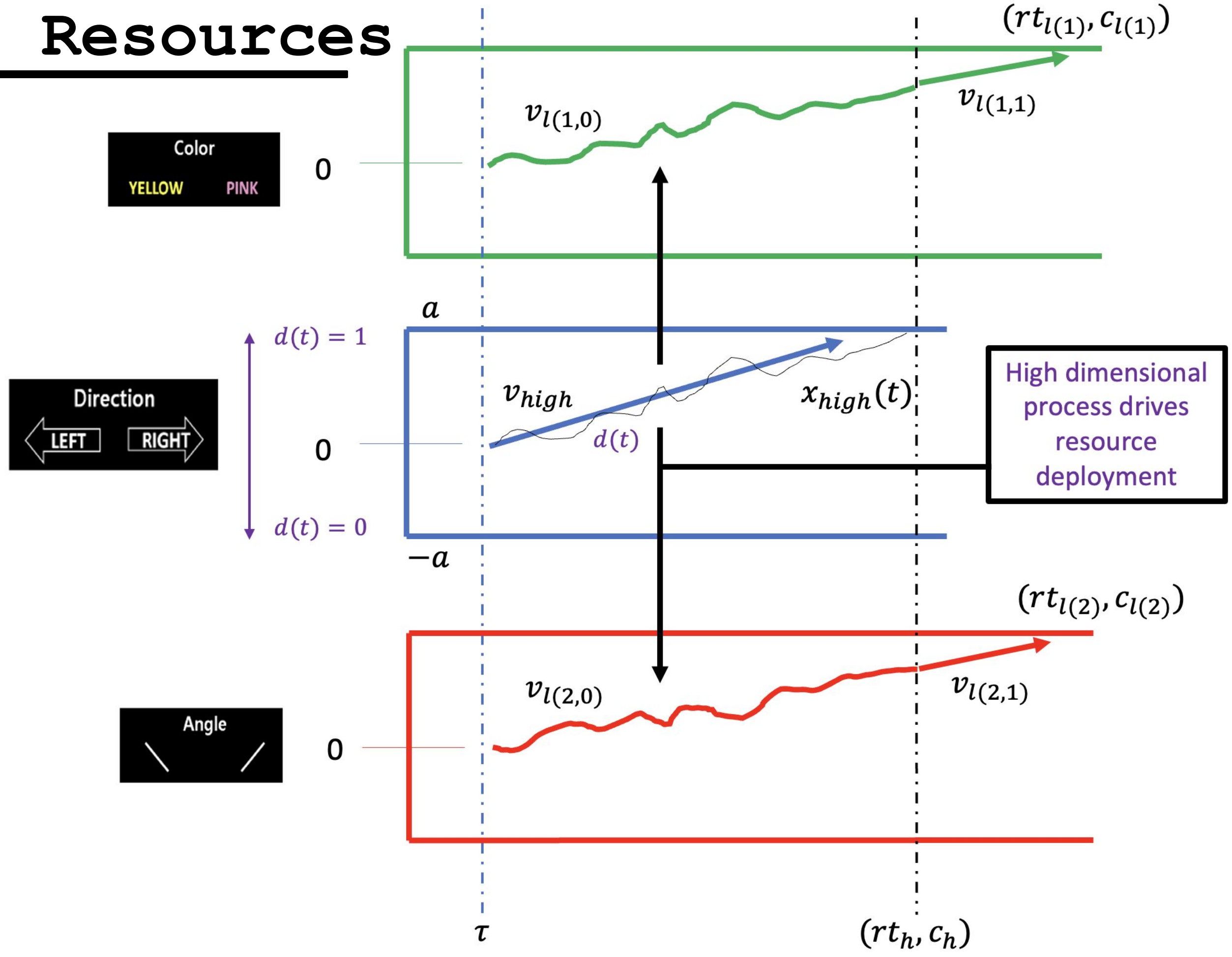
Hybrid: Finite Resources



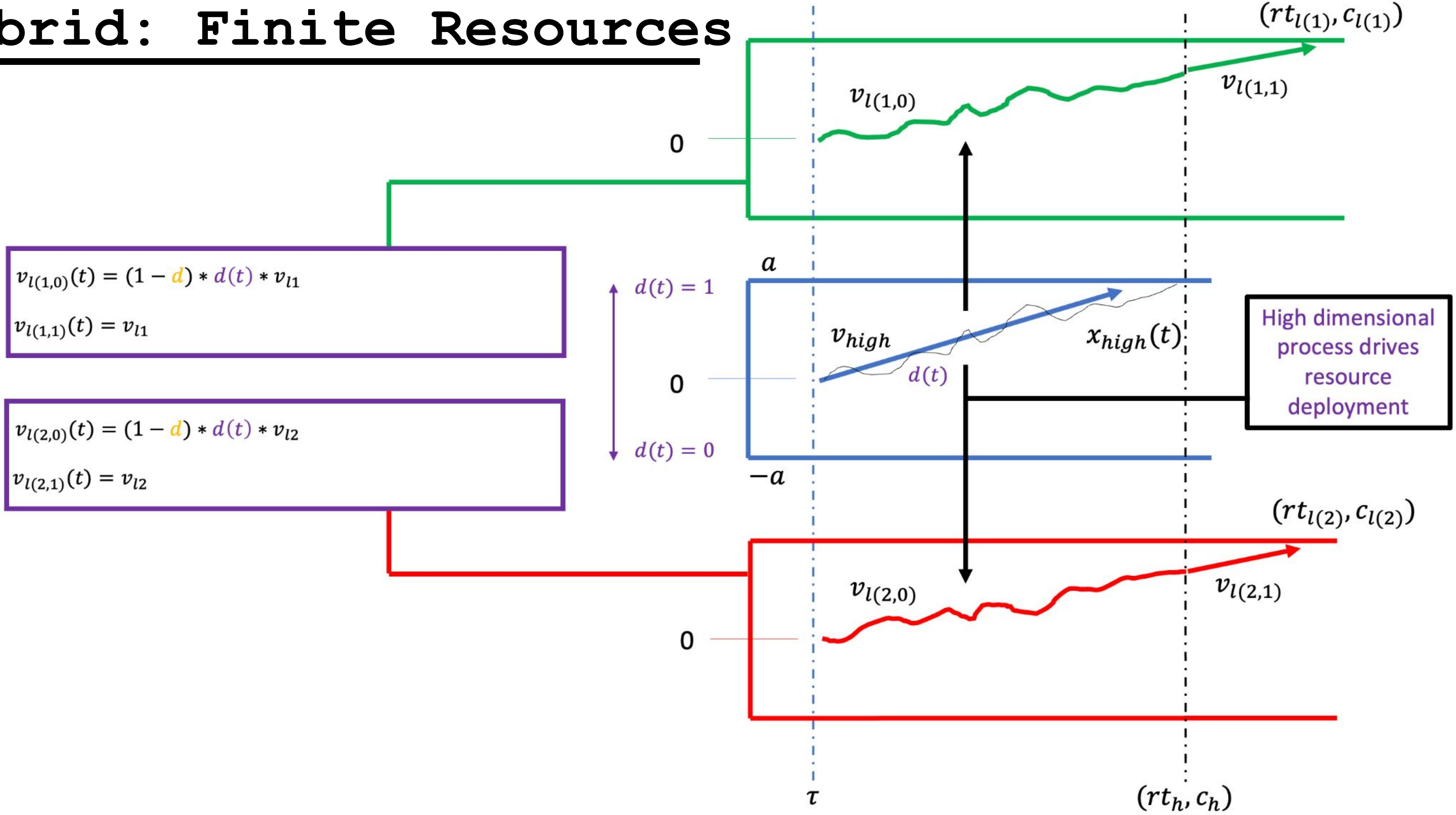


Color
YELLOW
PINK
0
$v_{l(1,0)}$
$v_{l(1,1)}$
$(rt_{l(1)}, c_{l(1)})$
Direction
LEFT
RIGHT
$d(t) = 1$
$d(t) = 0$
a
$-a$
0
v_{high}
$d(t)$
$x_{high}(t)$
Angle
0
$v_{l(2,0)}$
$v_{l(2,1)}$
$(rt_{l(2)}, c_{l(2)})$
τ
(rt_h, c_h)

Hybrid: Finite Resources



Hybrid: Finite Resources



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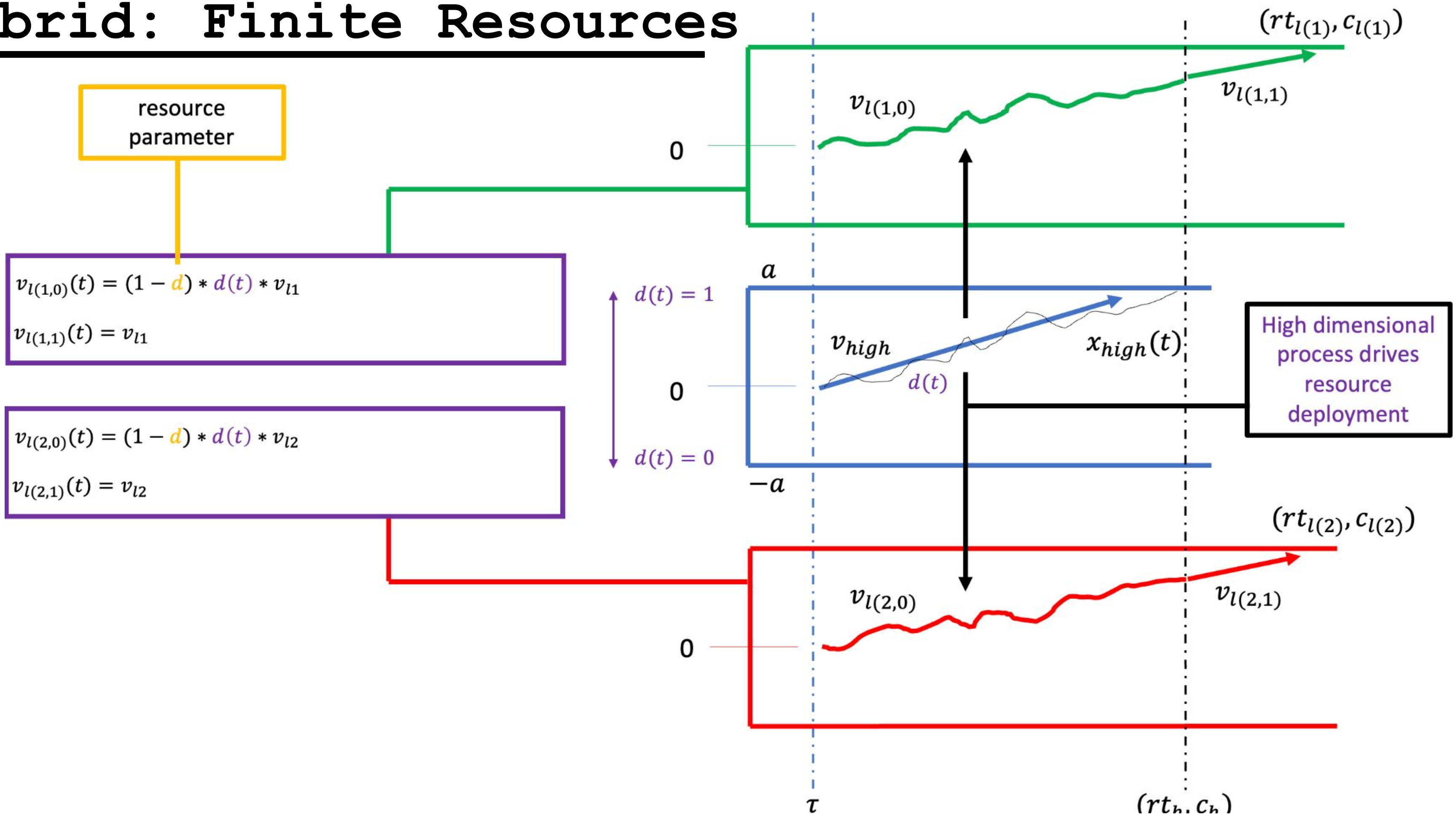
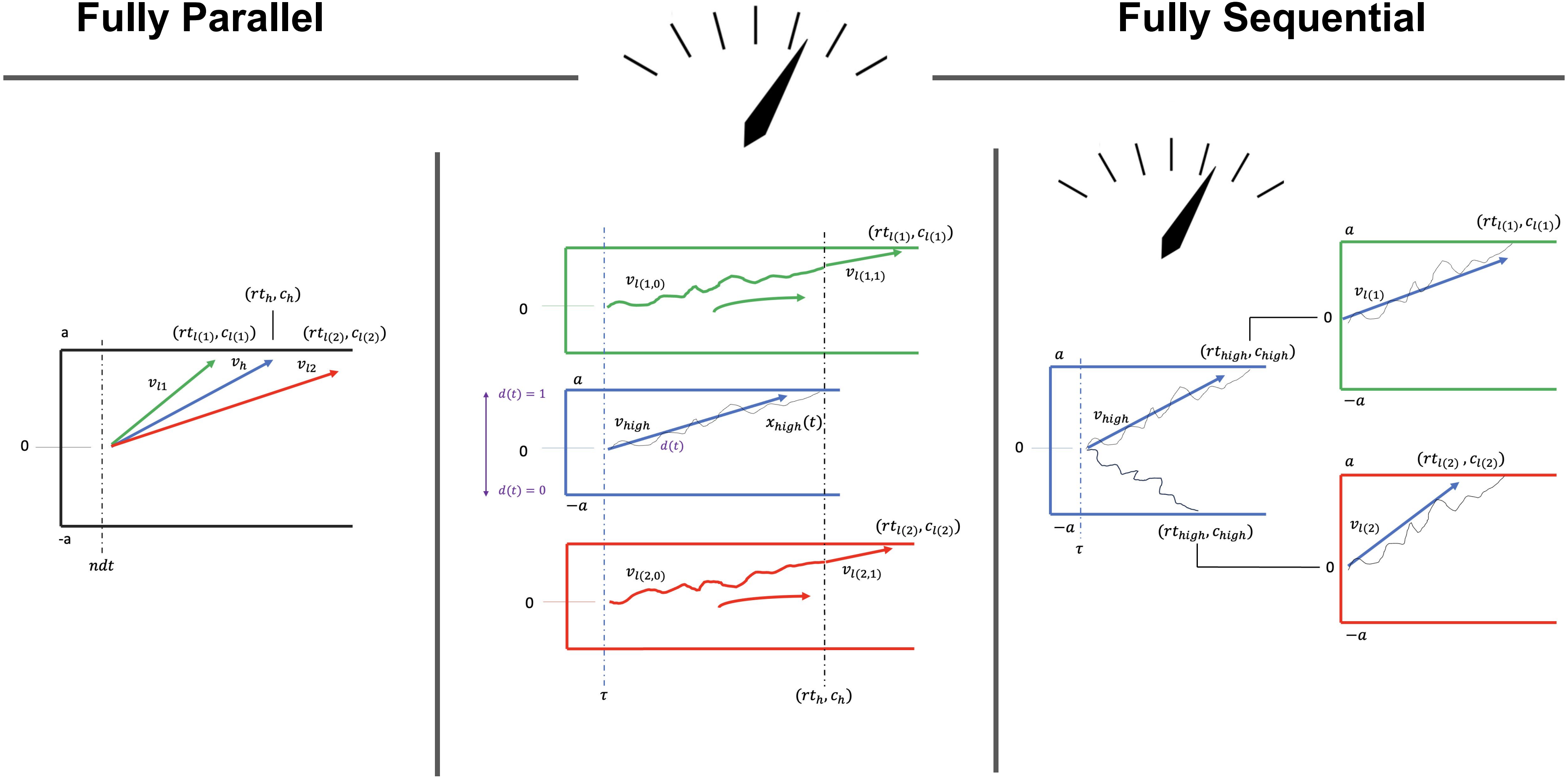
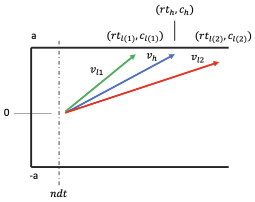


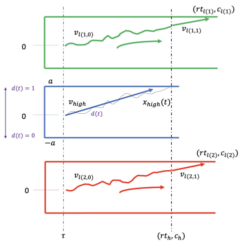
Fully Parallel

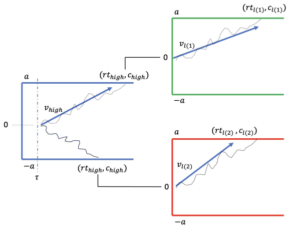
Fully Sequential



Strategy



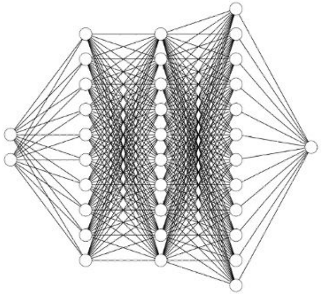




Forward Simulate

(Trivial)

Train LANs



ONNX

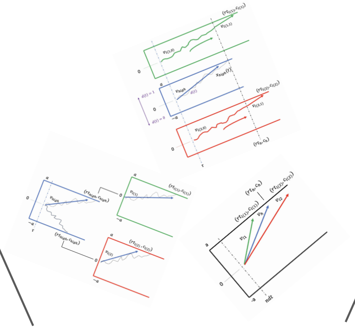
Toolbox

HSSM

$$p(\theta|x) \propto p(x|\theta)p(\theta)$$

- Hierarchies
- Trial-wise neural regressors by parameter
- etc.

Strategy

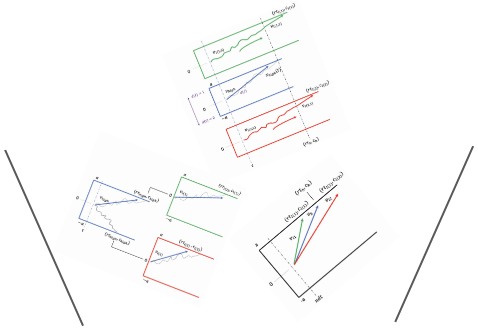


- Hierarchies
- Trial-wise neural regressors by parameter
- etc.

We have the networks in the toolbox, what next?

1. **Parameter recovery** studies on synthetic data mimicking structure of empirical dataset
   a. Simple fits
   b. Hierarchical models

Strategy



HSSM

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- Trial-wise neural regressors by parameter
- etc.

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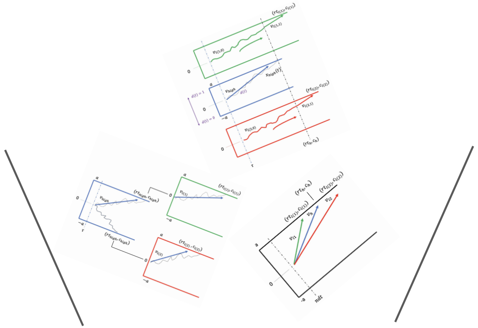
2. Analysis on **empirical data**
   a. Simple fits
   b. Hierarchical models
   c. Include trial-wise neural regressors (introduce EEG covariates)

Strategy



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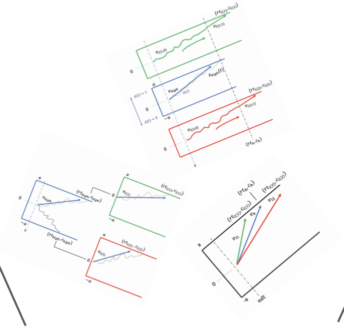
3. **Model Comparison**

Strategy



HSSM

- Hierarchies
- Trial-wise neural regressors by parameter
- etc.

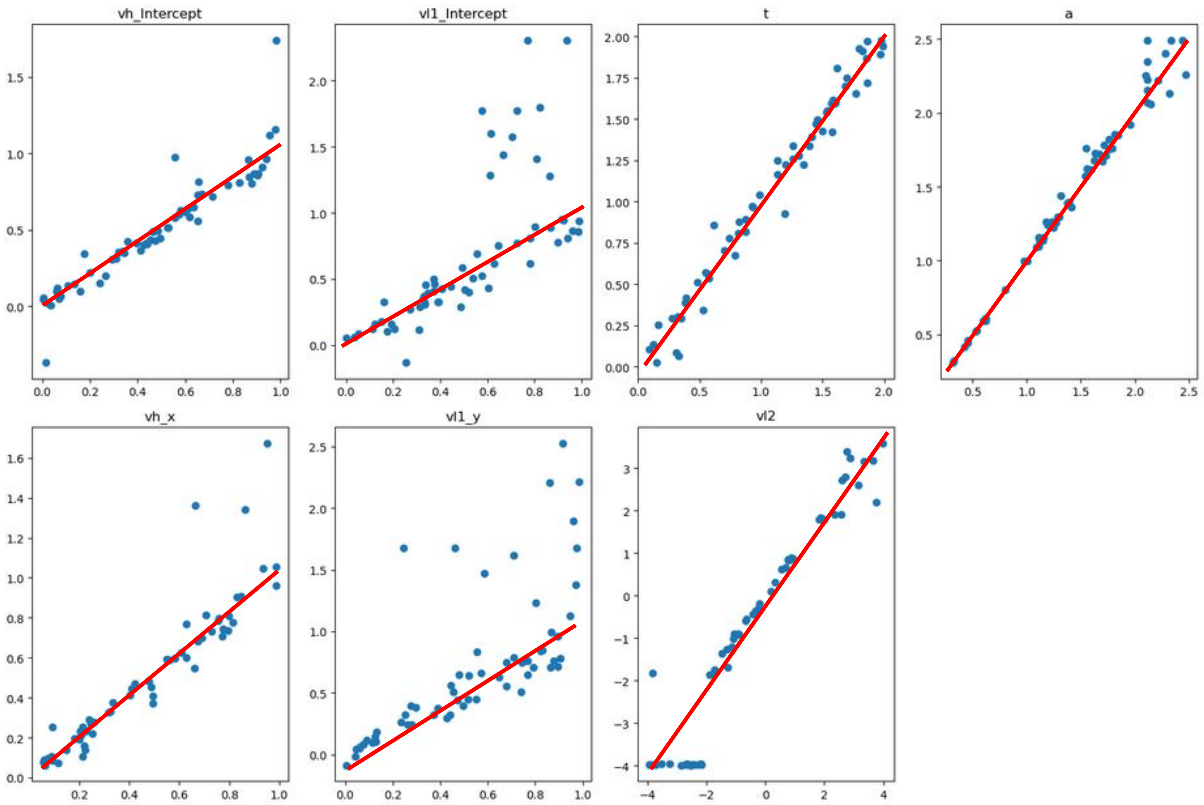
We have the networks in the toolbox, what next?

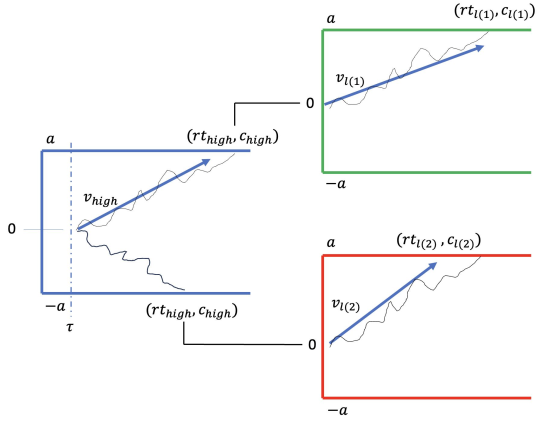
1. **Parameter recovery** studies on synthetic data mimicking structure of empirical dataset
   a. Simple fits → We are here
   b. Hierarchical models

2. Analysis on **empirical data**
   a. Simple fits
   b. Hierarchical models
   c. Include trial-wise neural regressors (introduce EEG covariates)

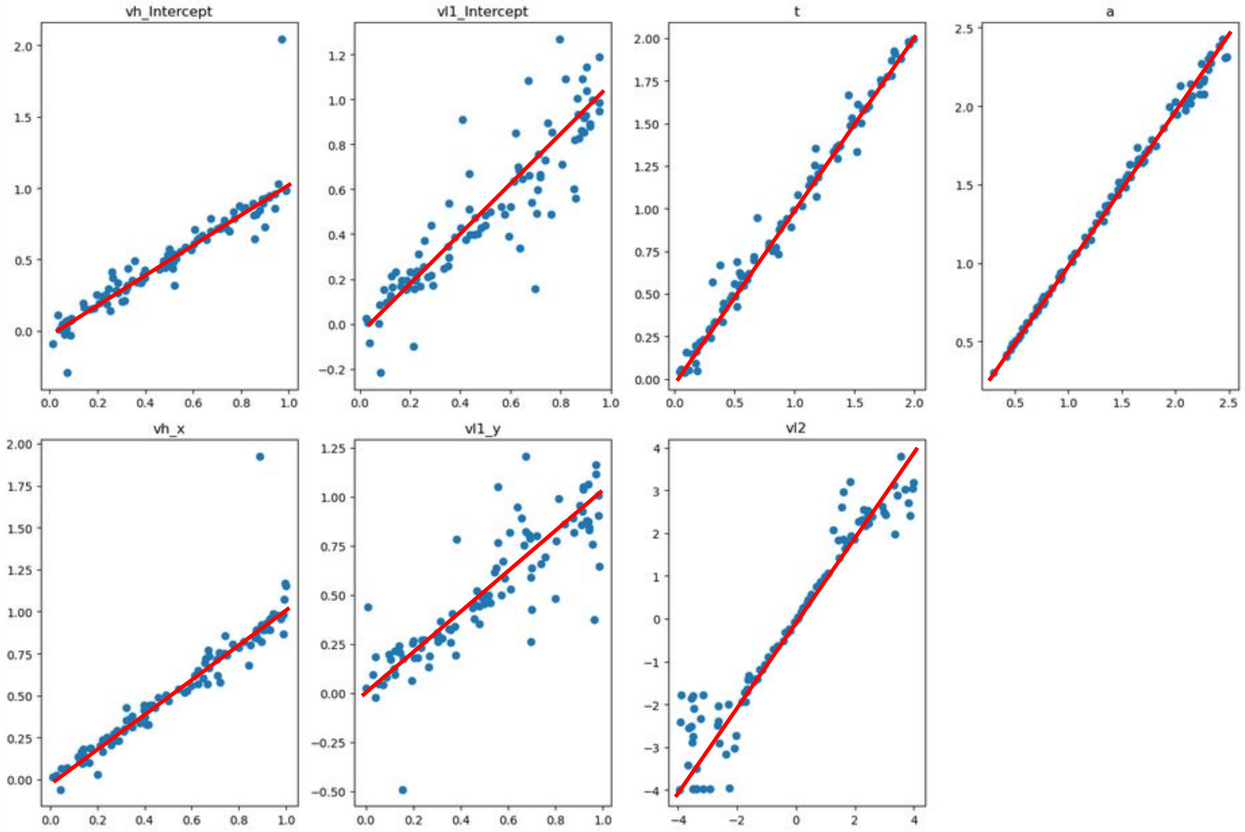
3. **Model Comparison**

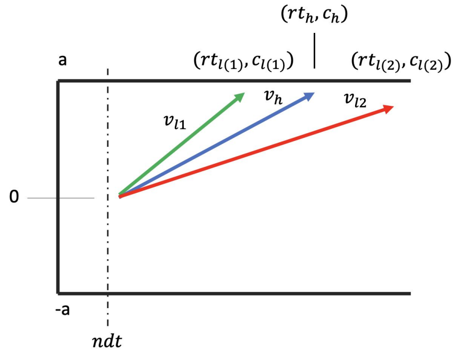
Parameter Recovery: Sequential



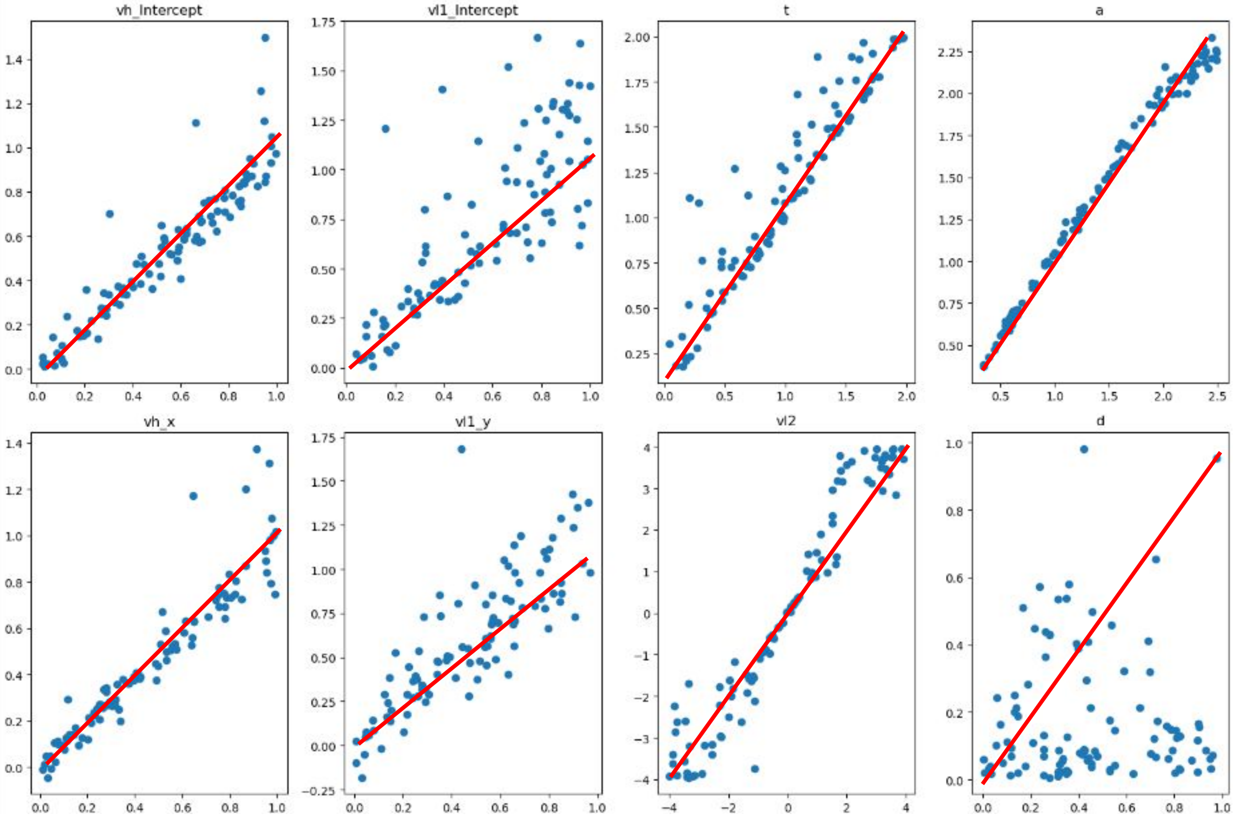


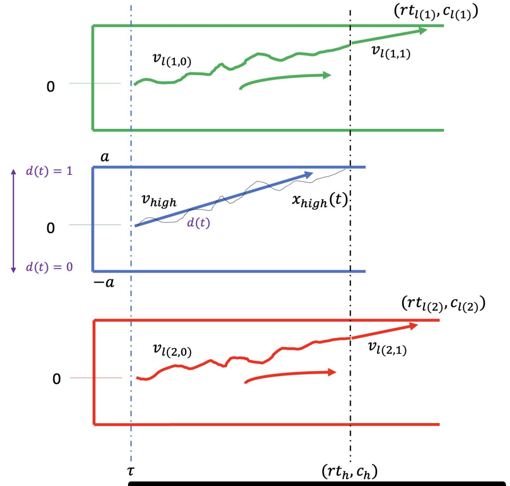
Parameter Recovery: Parallel





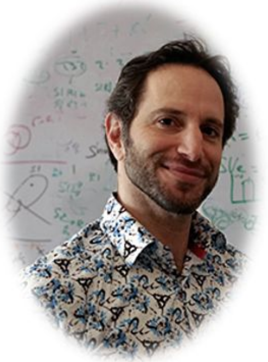
Parameter Recovery: Hybrid





Trouble identifying d parameter

End



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BRAIN
STORM

COGNITIVE LINGUISTIC & PSYCHOLOGICAL SCIENCES
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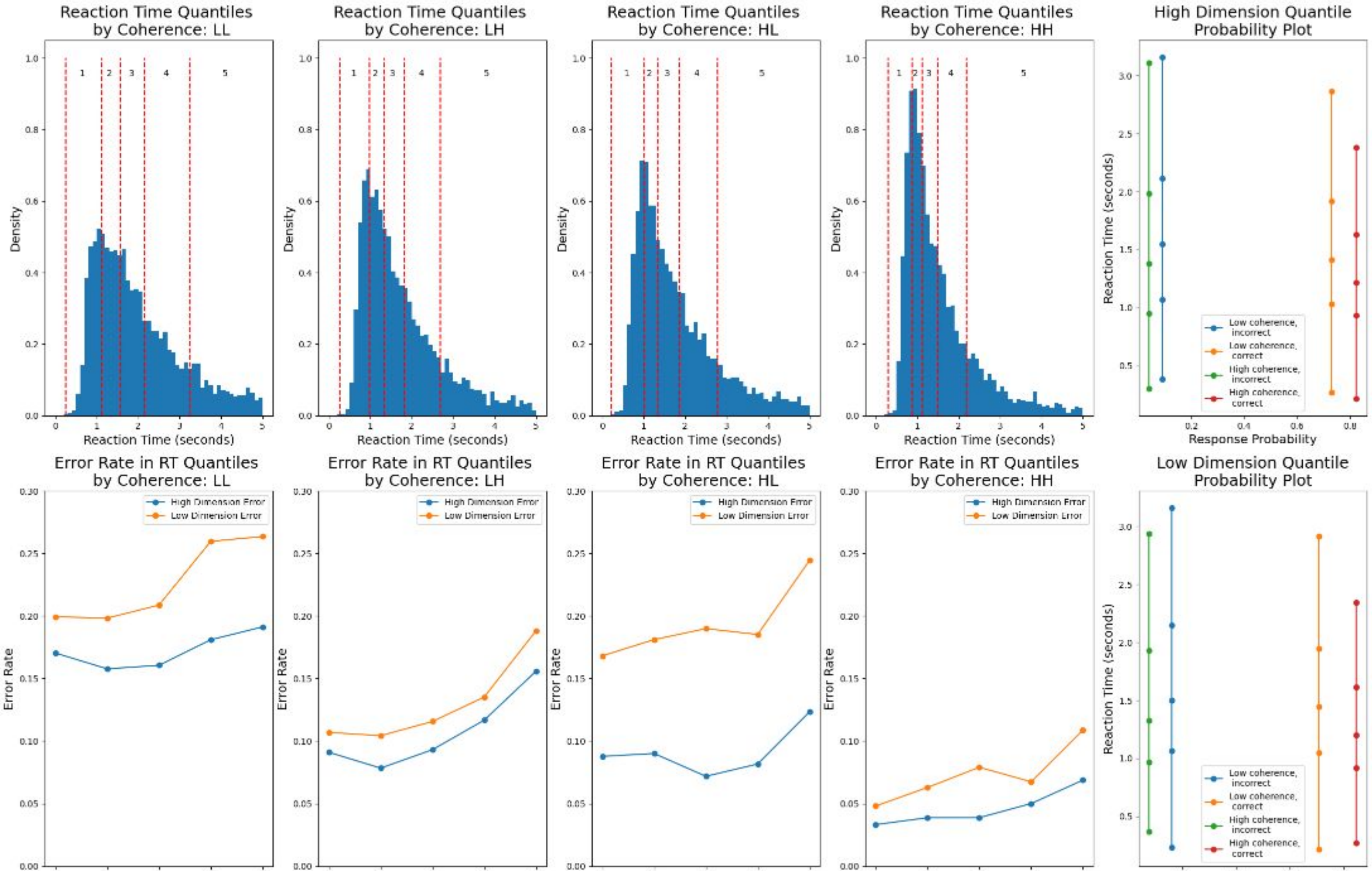
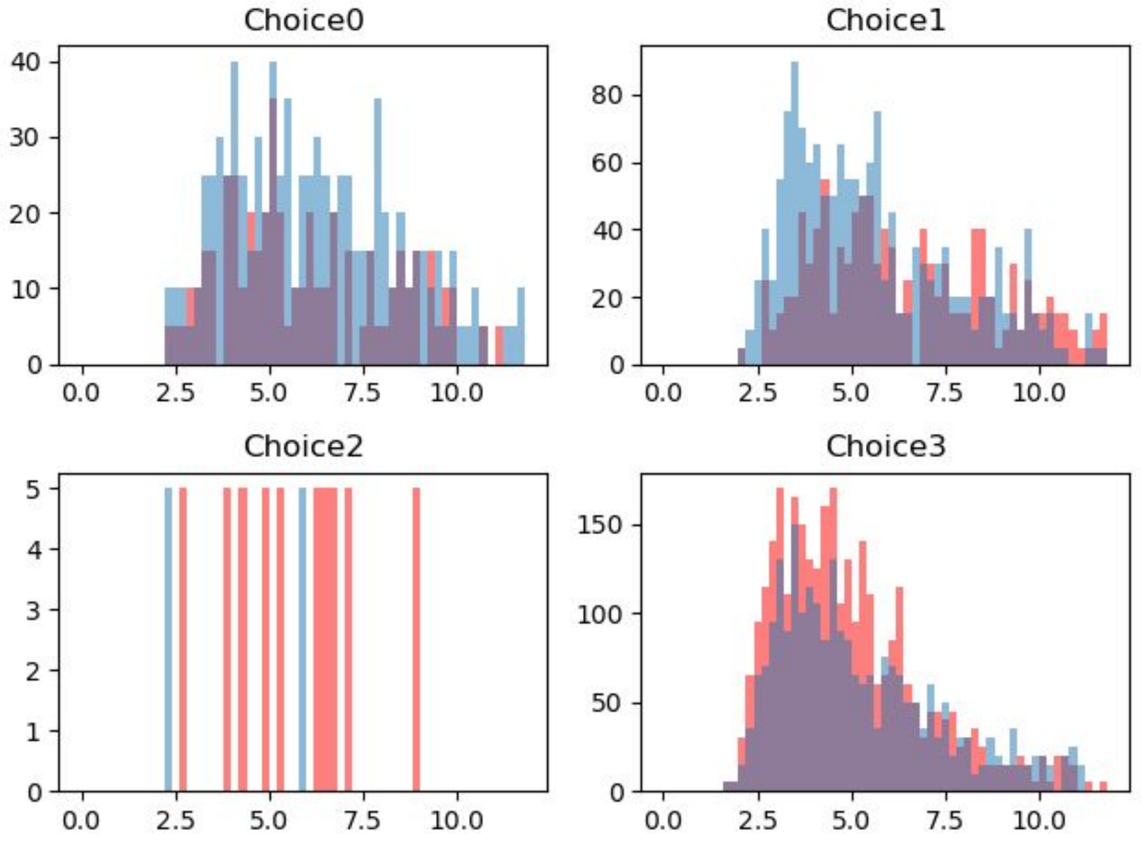
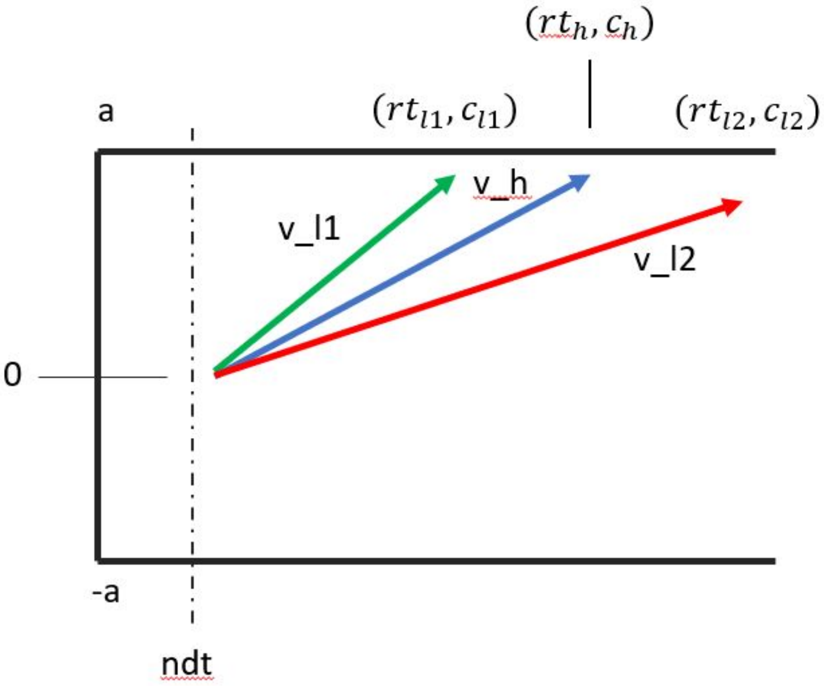
Observed Data Plot Bank



Seq 2 - Posterior predictive checks (RT vs. count)



Par2 Model



$$rt = \max(rt_h, rt_{l2}, rt_{l1})$$
$$c = (c_h, c_{l(c_h)})$$

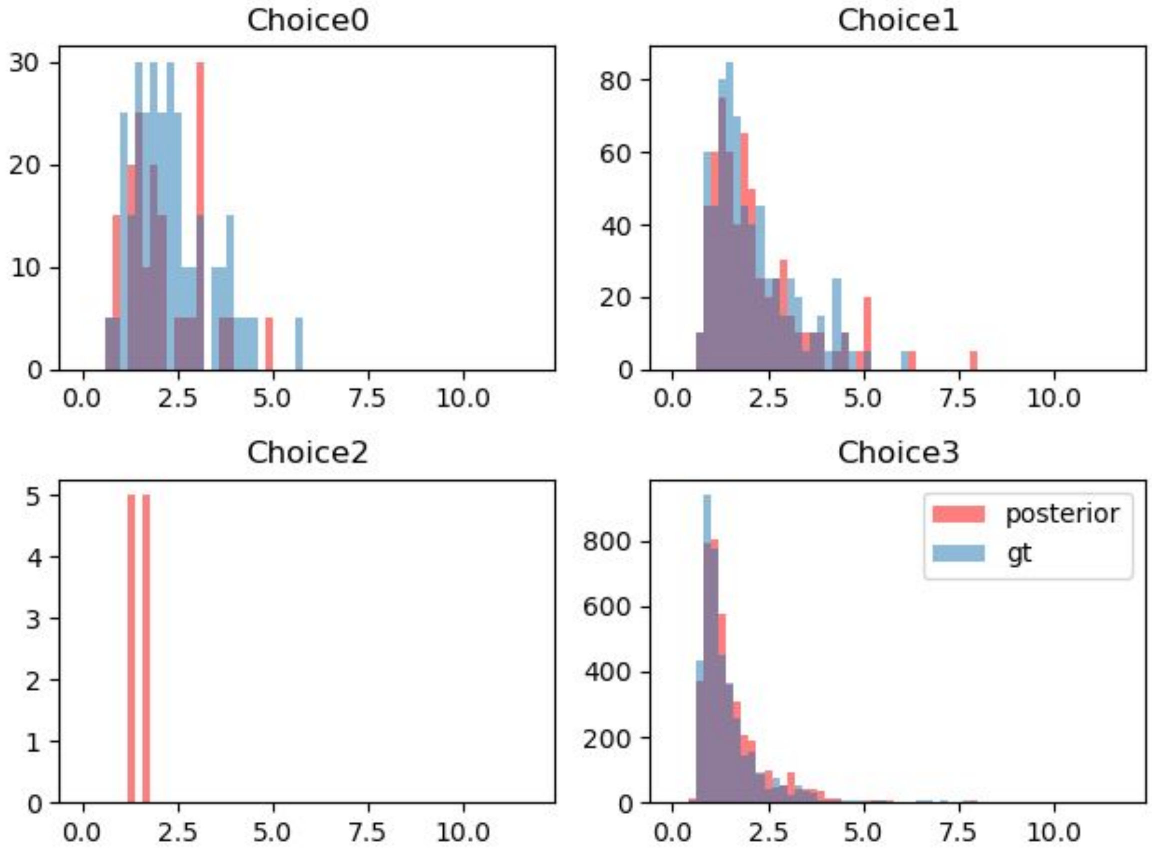
$$\Delta x_h(t) = \Delta t * v_h + \sqrt{\Delta t} * N(0,1)$$

$$\Delta x_l(t) = \Delta t * v_l + \sqrt{\Delta t} * N(0,1)$$

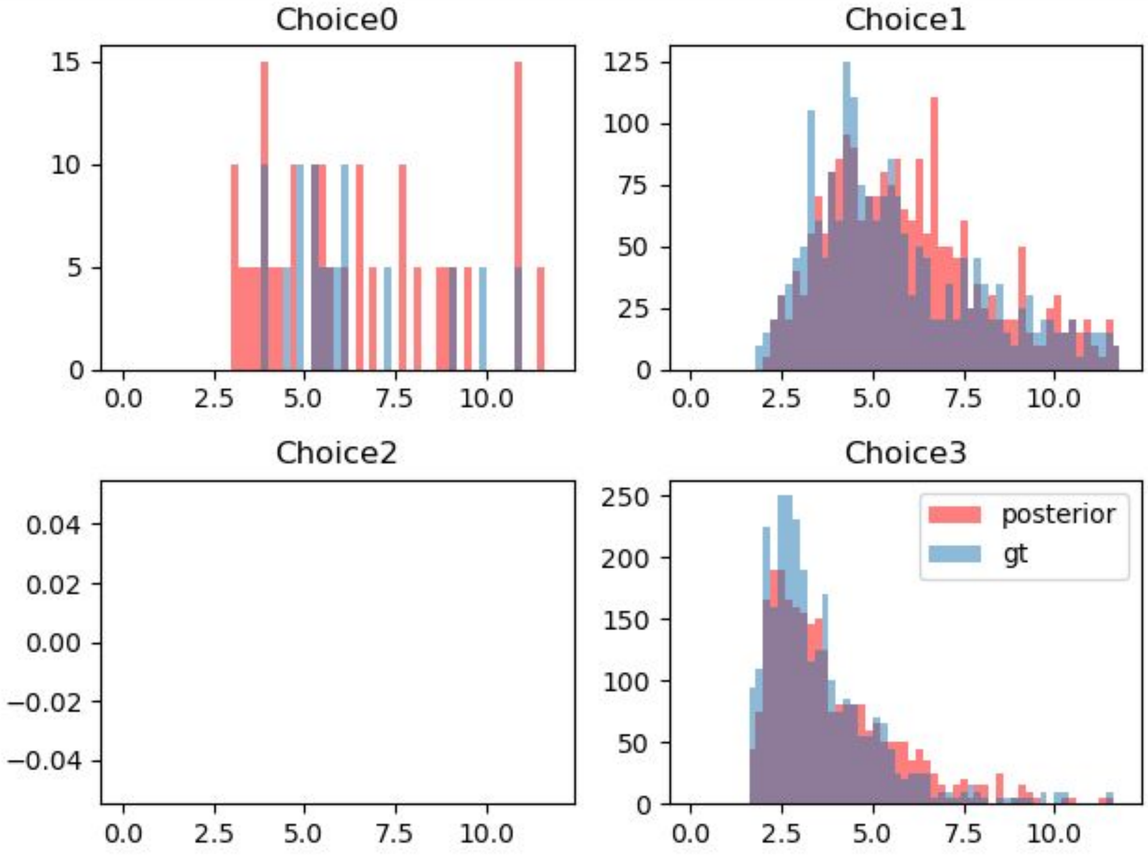
Parameters: $(v_h, v_{l1}, v_{l2}, a, \tau)$

SSMS: par2_no_bias

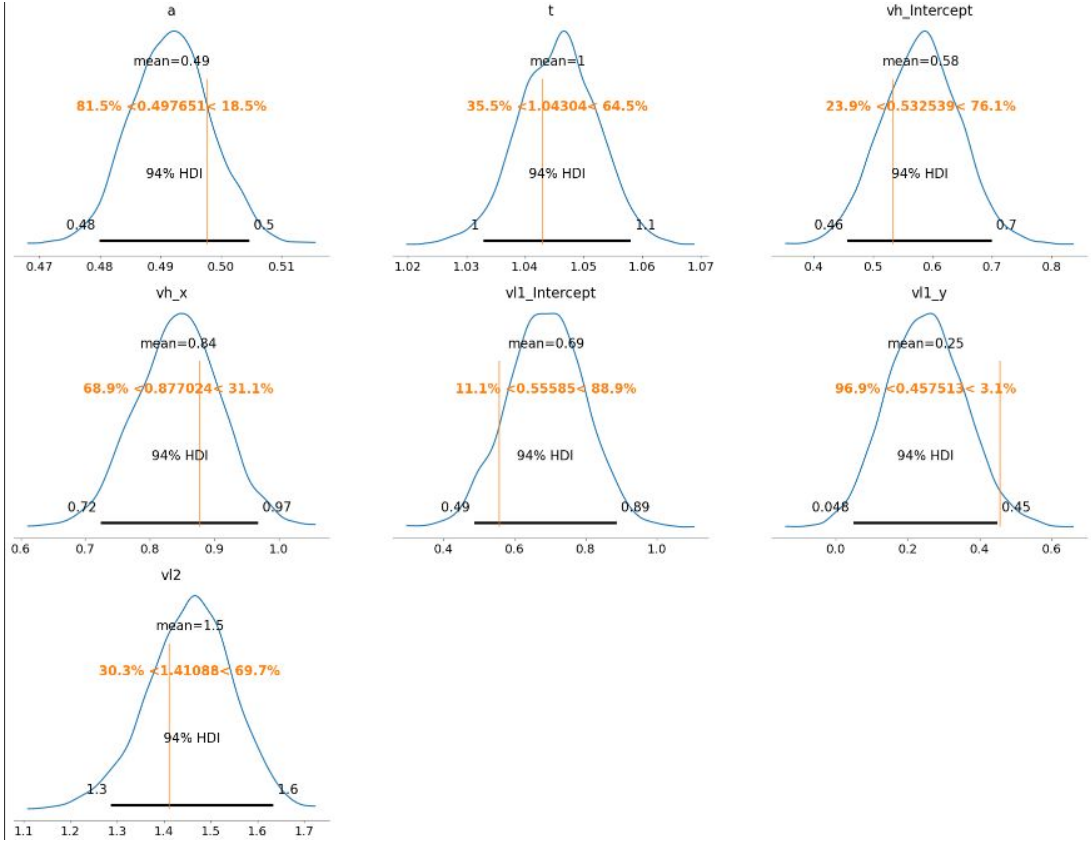
Parallel - posterior predictive check



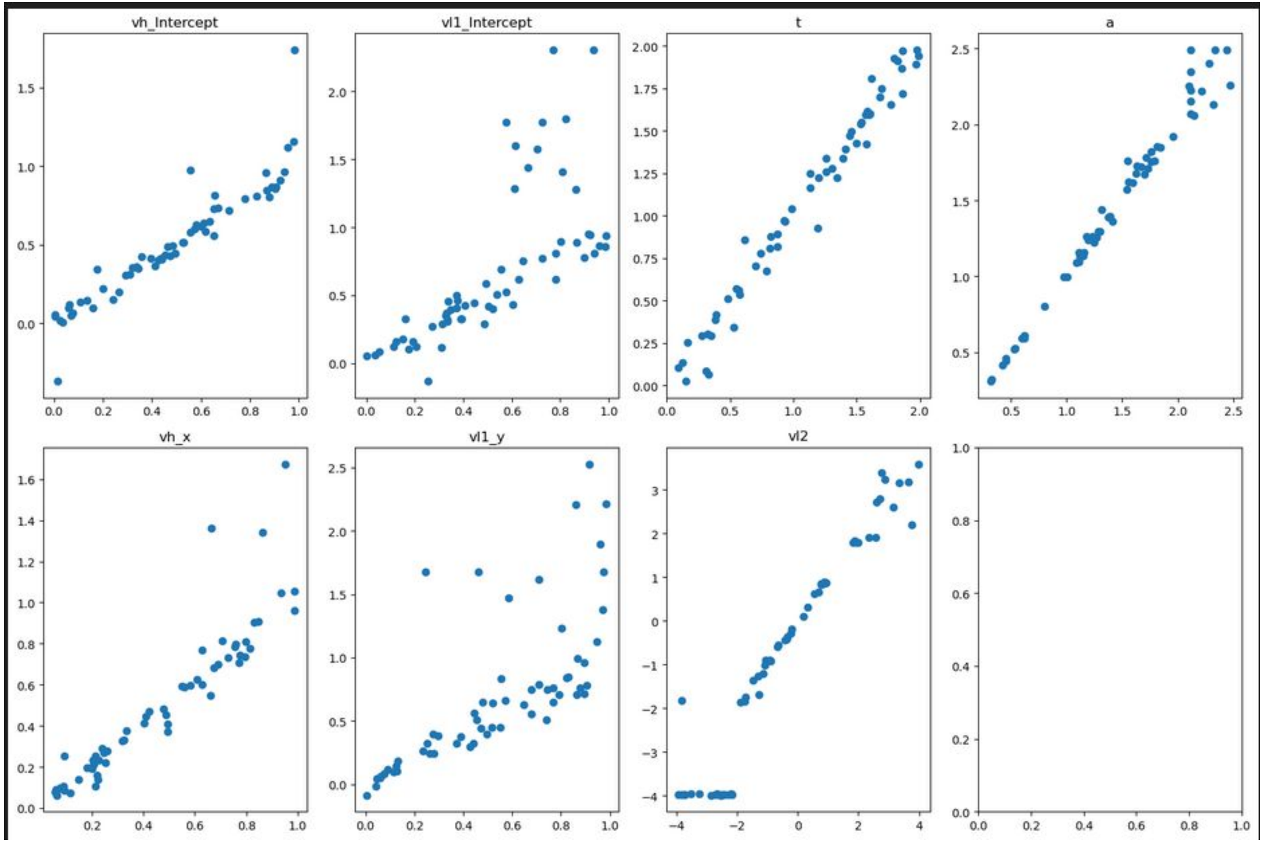
Leak - posterior predictive check



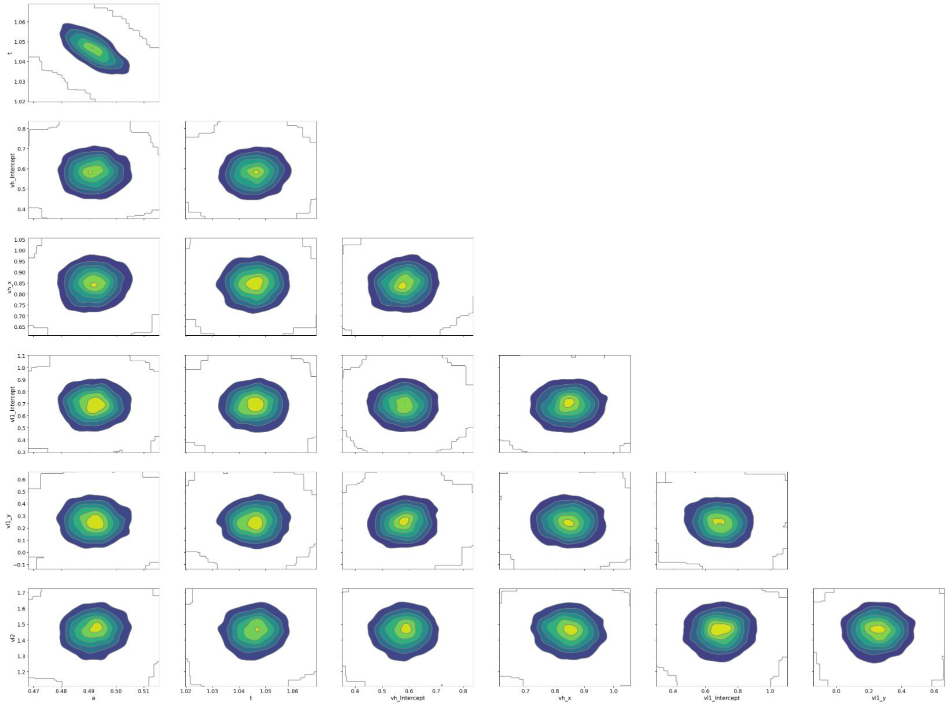
Single participant - synthetic data - sequential model



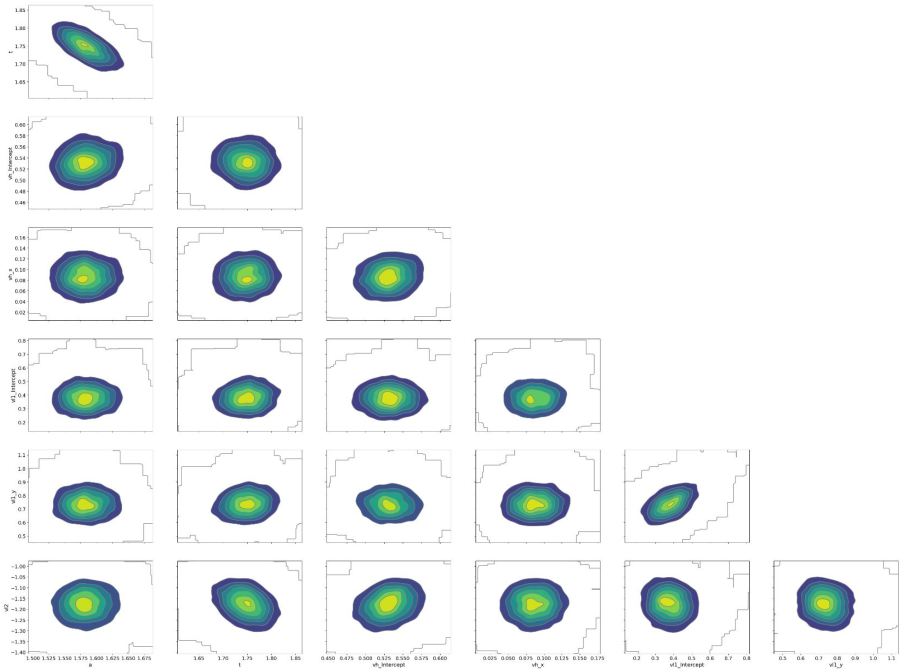
Parameter recovery - Sequential



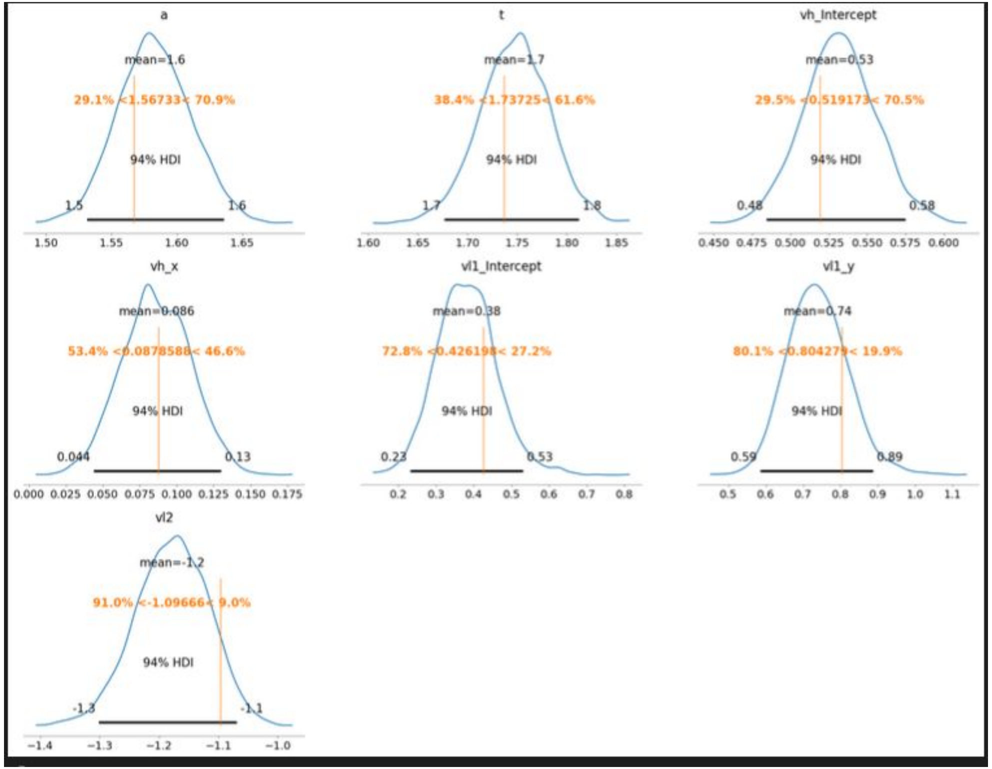
Single participant - empirical data

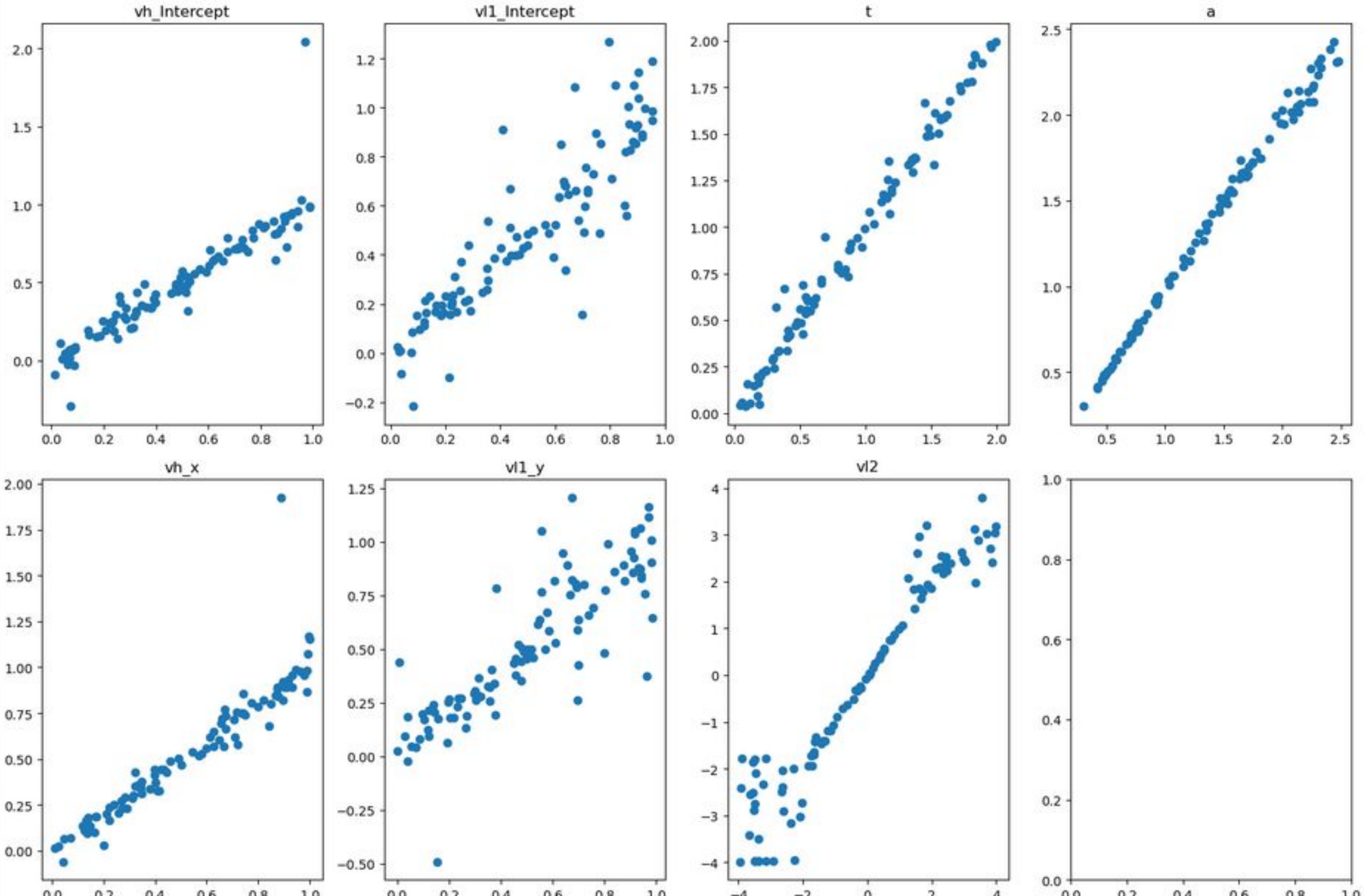


Single participant - empirical data



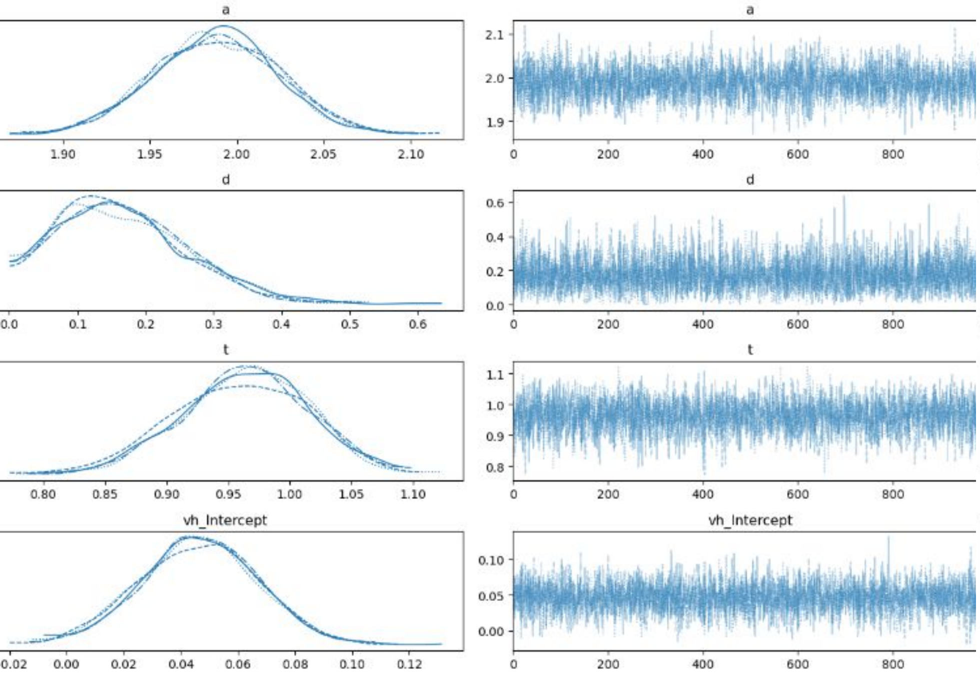
Single participant - synthetic data - parallel model

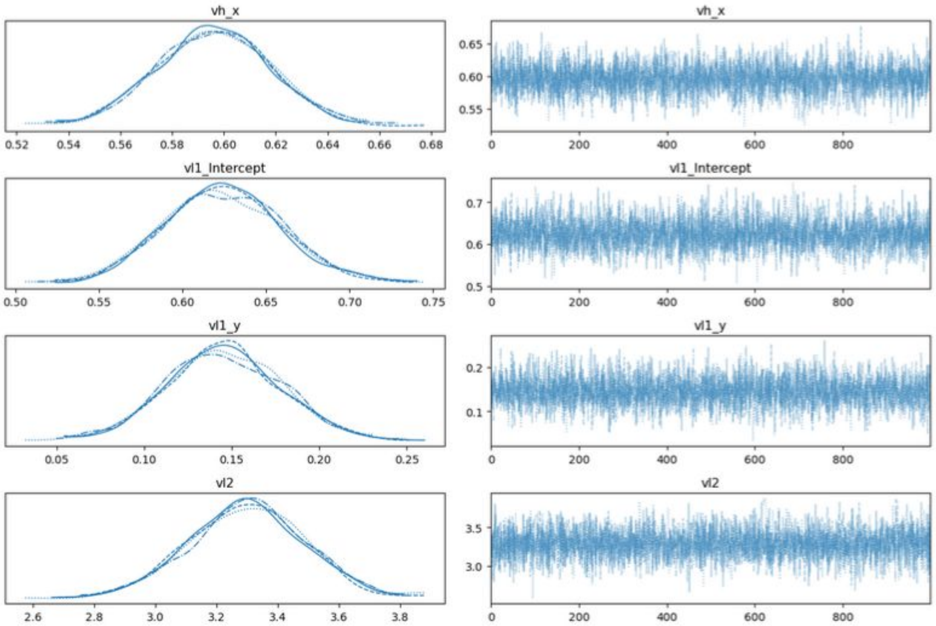




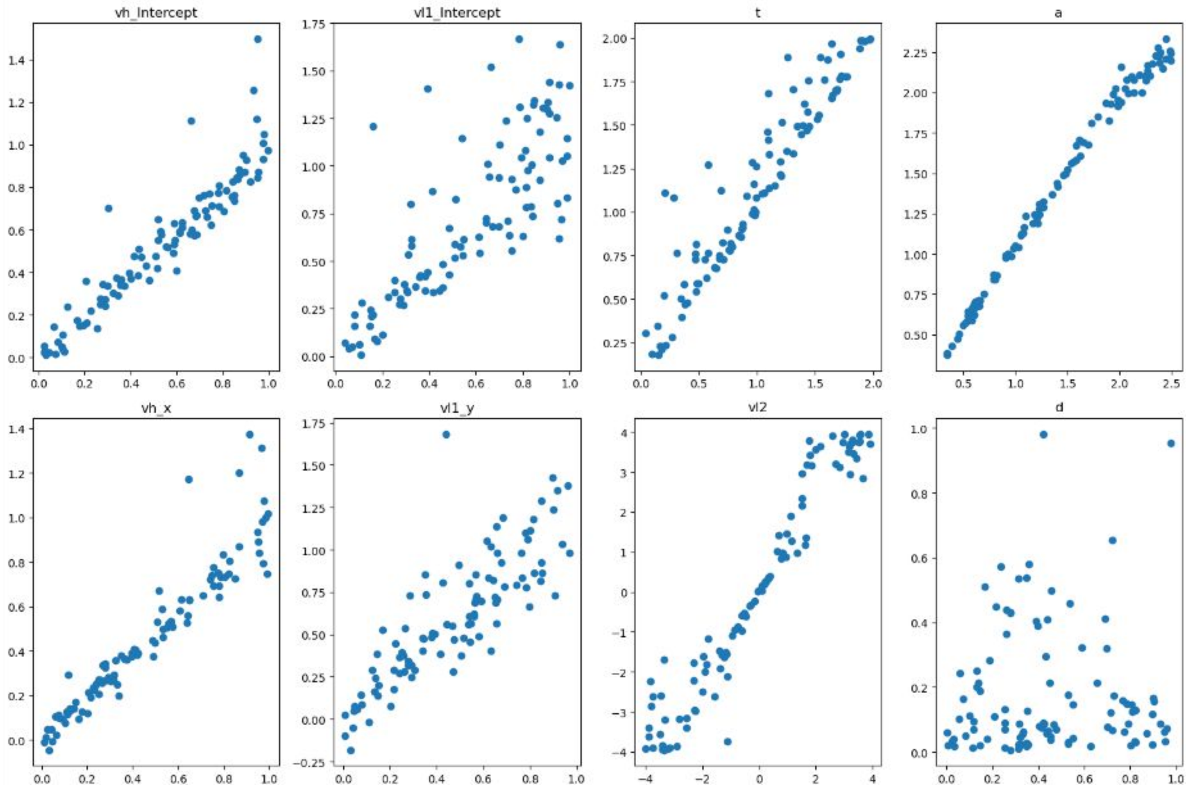
vh_Intercept
vl1_Intercept
t
a
vh_x
vl1_y
vl2

Single participant - synthetic data - leak model

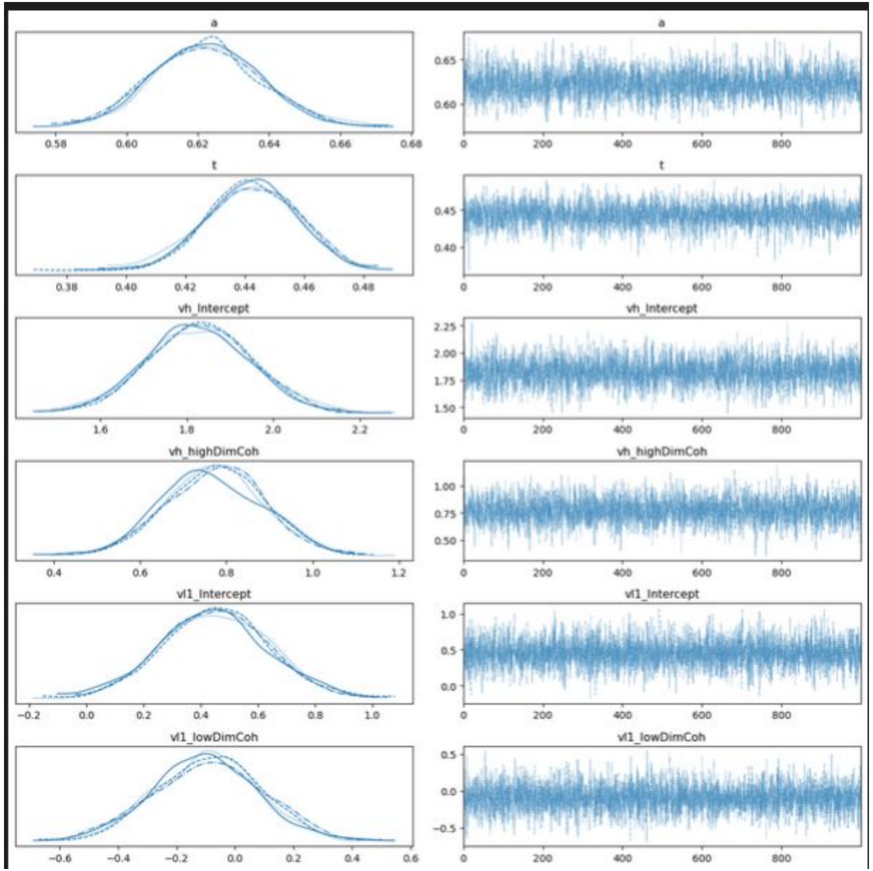


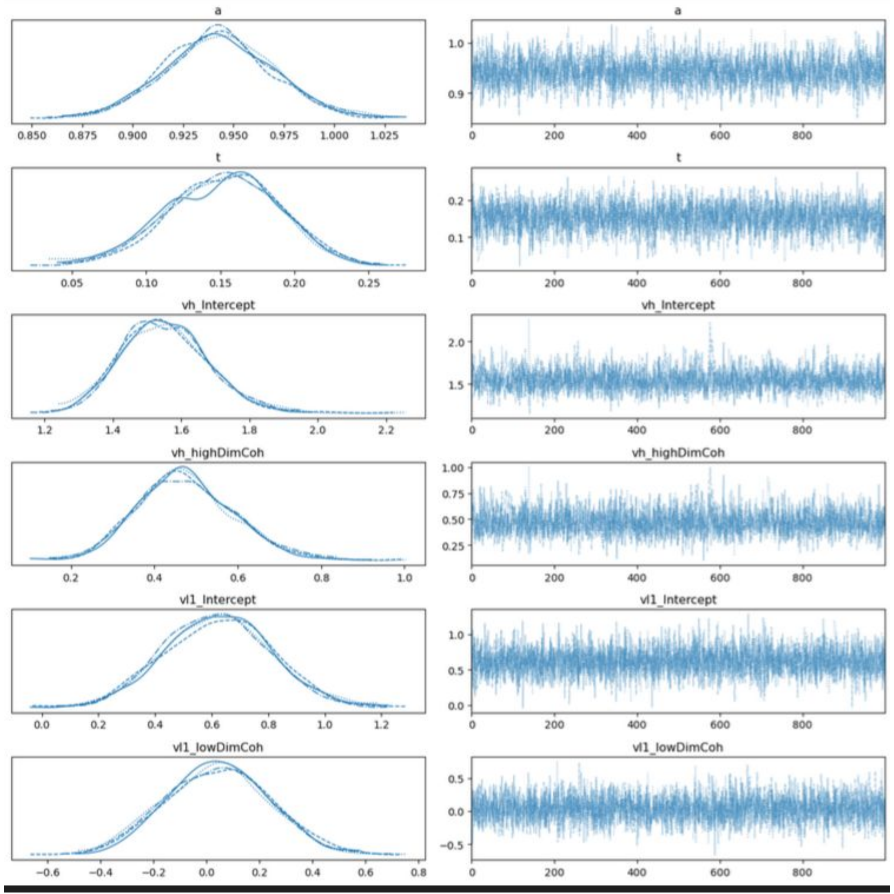


Parameter recovery - Leak

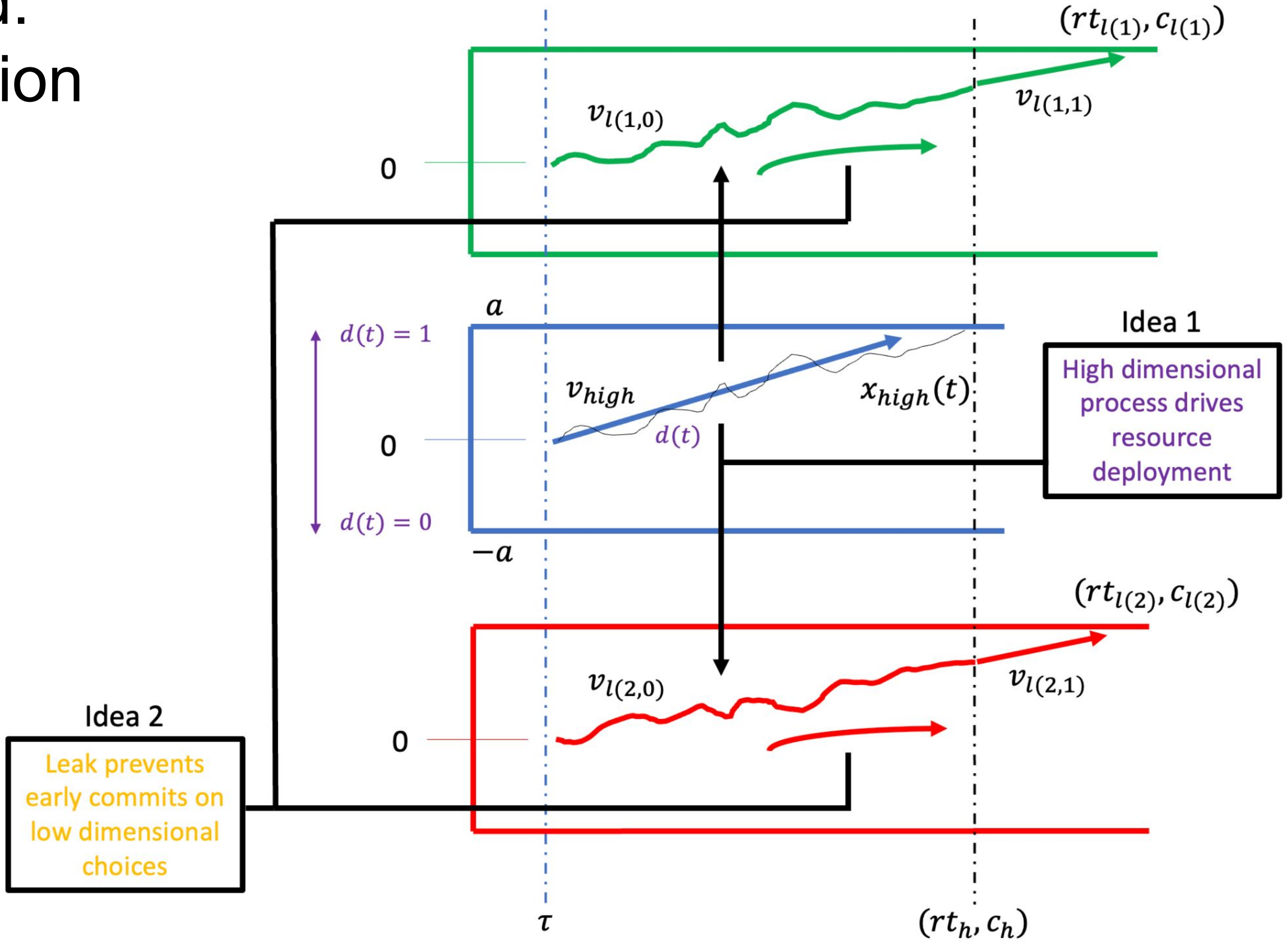


Empirical data - Sequential

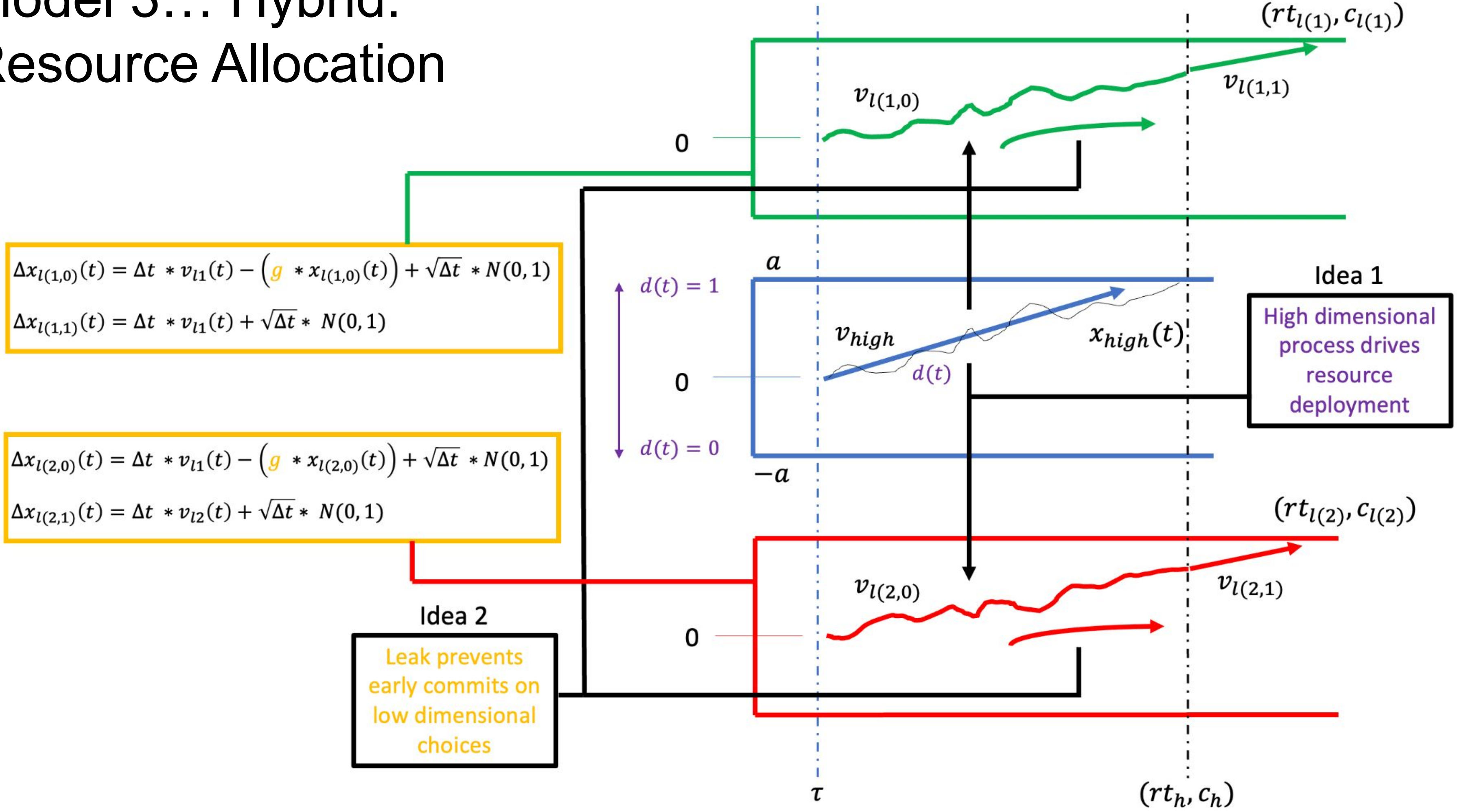




Model 3… Hybrid: Resource Allocation



Model 3… Hybrid: Resource Allocation



Model 3… Hybrid: Resource Allocation

$$\Delta x_{l(1,0)}(t) = \Delta t * v_{l1}(t) - \left(g * x_{l(1,0)}(t)\right) + \sqrt{\Delta t} * N(0,1)$$

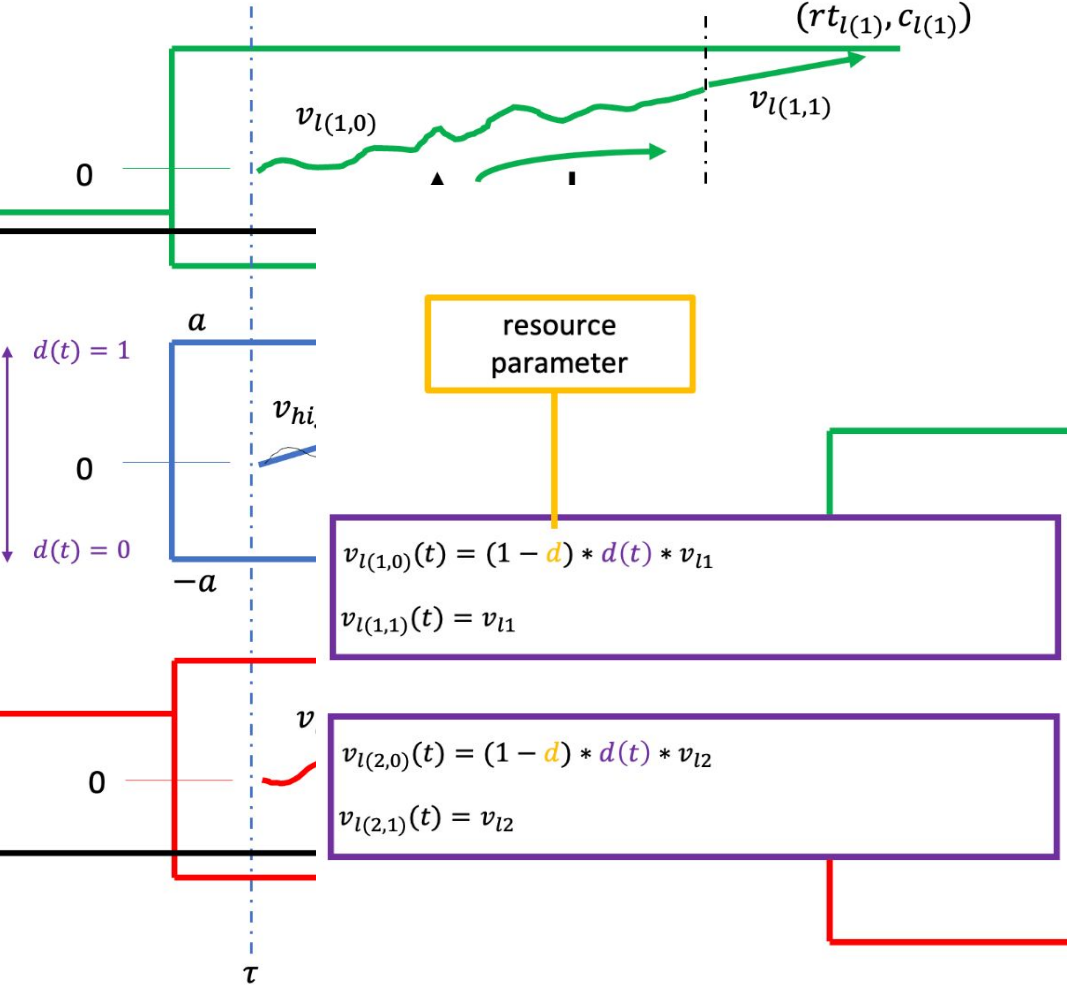
$$\Delta x_{l(1,1)}(t) = \Delta t * v_{l1}(t) + \sqrt{\Delta t} * N(0,1)$$

$$\Delta x_{l(2,0)}(t) = \Delta t * v_{l1}(t) - \left(g * x_{l(2,0)}(t)\right) + \sqrt{\Delta t} * N(0,1)$$

$$\Delta x_{l(2,1)}(t) = \Delta t * v_{l2}(t) + \sqrt{\Delta t} * N(0,1)$$

Idea 2

Leak prevents early commits on low dimensional choices



resource parameter

$$v_{l(1,0)}(t) = (1 - d) * d(t) * v_{l1}$$

$$v_{l(1,1)}(t) = v_{l1}$$

$$v_{l(2,0)}(t) = (1 - d) * d(t) * v_{l2}$$

$$v_{l(2,1)}(t) = v_{l2}$$